

BLOCK Box:

Developing a secure locker with parcel verification for blockchain integration



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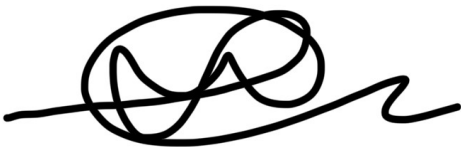
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Acknowledgements

No man is an island entire of itself; every man is a piece of the continent,
a part of the main.

—*John Donne*

Firstly, thank you to the Lord for the strength to persevere not only through this dissertation but my whole undergraduate degree.

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Furthermore, thank you to the music produced by NF and Drake! You two have my long study words an absolute ball!

To all my friends, this degree is not possible without you. Thank you.

To my family, my life and existence is not possible without you. Thank you. I love you all dearly.

Last but not least...

To LITUATION NATION...

‘CAN I GET A FIRE???’

I hope you all learn a thing or two about this revolutionary technology called blockchain. Happy reading!

Abstract

The logistics and supply chain industry has been completely transformed by the recent exponential growth of the e-commerce industry its the utilization of peer-to-peer marketplaces. Delivery is no longer just a linear service but is now competitive among providers in the realm of delivery speed, safety and sheer customer convenience. This demand for secure and efficient delivery solutions led to the development of this research project which is centered around developing a secure locker with parcel verification for blockchain integration. In essence, this project not only investigates the construction of a small secure package lock box but also the automation of the verification of the physical contents within the package. The added integration with the blockchain serves as a layer to enhance the overall security and operational efficiency of the system.

Traditionally, the parcel locker systems that have been in the market may use simple verification and access control methods such as keyed locks and manual package id inspections. However, these lockers are currently a problem as they are not scalable simply because more manpower is needed with the increase in delivery orders. Hence, this study then delves into modern locking mechanisms such as solenoid locks and wireless access control methodologies like NFC/RFID and QR codes. These not only enhance the security of the parcel locker system but also the overall user experience. However, the security is still not where it should be even with the use of these modern approaches which is where the integration with the blockchain comes in.

Blockchain technology offers an escrow solution where the concerns for pre-payment in peer-to-peer transactions are addressed. This solution is the powerhouse behind BLock Box's aim to solve the current ecommerce asymmetric payment problem of a party (usually the seller) having more transactional information. The technology is also capable of logging and storing all supply chain data that an off-chain arbitrator can refer to in cases where a buyer-seller dispute arises. An example of this data is the verification data that seeks to try and match the description of the item from the seller to the physical asset that the buyer has their hands on. This research delves into how this verification is made possible using advanced IoT weight sensing to categorise the item and then triggering a Blockchain transaction should the verification process be successful. The project goes deeper and provides the state of the system in real-time through a public JSON formatted endpoint for future Blockchain smart contract interactions.

Nevertheless, the world is currently being dominated by peer-to-peer transactions with people taking a lot of time and money on online ecommerce platforms. This research about the BLock Box is, therefore, a presentation on how to solve the asymmetric payment problem within economic peer-to-peer transactions in a parcel locker last-mile delivery framework.

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Chapter 1

Introduction

1.1 Background to the study

The state of the logistics and supply chain industry is unrecognisable as of late partly because of the exponential growth of the e-commerce industry. As online shopping has increasingly become popular, the millions of parcels being delivered everyday has induced the natural demand for secure and efficient delivery solutions, particularly in the last-mile. The last-mile refers to the very last step of a supply chain delivery process where a parcel is handed over to the customer. This stage is often coined to be the most costly because of issues such as missed deliveries.

One remedy to trying to diffuse this high demand for delivery services is the development of secure parcel locker systems. These robust, innovative lockers provide a safe and efficient mechanism for customers to receive their packages at a time that suits them. Moreover, apart from alleviating issues related to missed deliveries this solution is also immune to the current porch package theft trend. Companies such as Amazon, DSV and Pudo have already implemented parcel lockers to a great extent, particularly in densely populated urban areas.

However, the only noticeable caveat of this logistics system apart from the slight inconvenience of a customer having to get their package themselves is that it is still highly dependent on a lot of trust from all the parties involved. In cases of paying for the item before receiving it, the buyer has to trust the seller to deliver the exact item they ordered. This need for trust can pose an immense challenge particularly in peer-to-peer marketplaces like Facebook Marketplace, where buyers and sellers are typically strangers. Furthermore, the seller has to trust that the personnel handling the delivery will indeed place that specific item in the correct locker should the seller not be doing it themselves. These are just a few examples of the elements of trust required to make a parcel locker delivery system work.

Hence, the integration of smart contracts from the realm of blockchain technology with the parcel locker system is a way of making the system inherently trusted and efficient. These smart contracts automate secure transactions using a set of parameterised contractual agreements set by the involved parties. Essentially the funds are locked in escrow, thus, removing the need for a trusted third-party and then released upon the verification of the underlying traded asset. This integration of blockchain technology essentially also solves the famous economical theory of transactional asymmetric information with the mentioned escrow service. This is because the automated verification process of the buyer and seller ensures that the buyer and seller have the same level of information regarding the product before the actual transaction takes place. The only complexity with the actual asset verification implementation of this is that physical assets do not integrate easily with blockchain technology as compared to their digital counterparts. However, advances in IoT technology, Web3 frameworks, computer vision and machine learning have the potential to seamlessly automate the full verification of physical assets.

1.2 Objectives of this study

The objective of this study is to investigate and develop a secure locker system named BLock Box that solves the economic asymmetric product information problem for peer-to-peer transactions. This system uses modern IoT technologies to address various last-mile delivery inefficiencies to be able to solve the underlying problem

1.2.1 Problems to be investigated

This investigation will focus on several key questions:

- What is the asymmetric information theory and its place within this investigation?
- What are the existing challenges with parcel locker systems and where are they currently in use?
- What are the operational and security inefficiencies with these lockers and how are they currently being addressed?
- How would the security and operational efficiency of these lockers be enhanced with AI and blockchain integration?

1.2.2 Purpose of the study

The purpose of this study is to investigate the integration of blockchain technology and IoT technology in solving the asymmetric peer-to-peer transaction information problem by creating of a safe and smart last-mile delivery parcel locker. Therefore, the prototype illustrated in the study is a sample of a system that would help make the parcel delivery service industry safer, more effective, and easier to use for the end-user. This is particularly important in an era where online shopping is the norm and customer expectations for dependable, consistent delivery services are on the rise.

1.3 Scope and Limitations

The underlying research in this report focuses on the design and implementation of a parcel locker system as a last-mile delivery solution. It is imperative that the scope of this report clearly be limited to the last-mile as the logistics and supply chain industry is very broad with a wide range of processes. Therefore, the ways in which blockchain technology can improve the parcel locker last-mile delivery process is the focus of this report.

Moreover, it delves into the actual implementation of a public transactional smart contract that would be used to deposit the funds from the buyer to the seller once the parcel contents are verified. However, this project does not go into the actual implementation of developing external adapters that can connect the off chain sensor data to the public chain. The adapters stage is simply bypassed and the blockchain transaction is developed using all the local data and parameterised on a state to state level. This is because the emphasis is more on discussing the feasibility of integrating the blockchain technology with IoT technology to solve the asymmetric information problem by ensuring that a blockchain transaction can be triggered purely with an IoT based system. Ideally, one would have a physically decentralised system as well with multiple microcontrollers and sensors so that that the system is robust but this would introduce an embedded system design complexity and also exceed the R2 000.00 budget constraint of the project so the prototype discussed has a centralised framework.

1.4 Plan of Development

This report is organized into ten chapters. This chapter introduced the project, outlining the background and objectives. Chapter 2 details the system requirements. Chapter 3 provides the theoretical background of blockchain technology, followed by Chapter 4, which reviews relevant literature and existing solutions. Chapter 5 describes the design and implementation of the BLock Box system. Chapter 6 outlines the testing and validation procedures. Chapter 7 presents the testing outcomes, and Chapter 8 analyzes and interprets these results. Chapter 9 then concludes the investigation with Chapter 10 offering recommendations on improving BLock Box.

Chapter 2

Requirements

2.1 Requirements and specifications

2.1.1 User Requirements

A product is only as good as the overall satisfaction it provides its end-user. Hence, the user requirements for the proposed BLock Box system are as follows:

Table 2.1: BLock Box User Requirements

UR-ID	BLock Box User Requirements
UR-01	The user needs a secure parcel locker system to store small parcels.
UR-02	The system should be reliable and always be operational unless power or component failure which it must then stay locked until a technician is able to come fix the issue.
UR-03	The access needs to be limited in that the user needs a certain level of authority in order to access a parcel inside BLock Box.
UR-04	The locking mechanisms should be keyless.
UR-05	The system must be able to integrate with smart contracts and blockchain technology.
UR-06	Feasible integration with peer-to-peer marketplaces.
UR-07	The system must be cost effective.
UR-08	The security of the physical locker should be robust and be able to resist basic burglary attempts.

2.1.2 Requirement analysis

The requirement analysis for the BLock Box system can be separated into functional requirements as well as system specifications. These are tabulated in the corresponding tables, [Table 2.1](#) and [Table 2.2](#). However, it is imperative to mention that functional requirements are different to user requirements in that they focus more on breaking down those aspects that the system has to accomplish.

Functional Requirements

Table 2.2: BLock Box Functional Requirements

FR-ID	BLock Box Functional Requirements	UR fulfillment
FR-01	The lock system needs to be able to store an international standard small package size.	UR-01
FR-02	The access to the lock system must be efficient, timed, temporary and allowed only with the right authorisation.	UR-02 UR-03
FR-03	The lock system must be a fail-secure electronically controllable locking mechanism for efficient access control.	UR-02 UR-03 UR-04
FR-04	The system needs to handle personal information securely.	UR-03
FR-05	The system has to incorporate an electronic sensor to verify parcel contents.	UR-04
FR-06	There needs to be a user interface system integrating the all parties of a peer-to-peer transaction.	UR-06
FR-07	The system needs to take the IoT sensor data and use it to trigger a blockchain-based transaction.	UR-05
FR-08	The system needs to be connected to the internet for further blockchain development.	UR-05
FR-09	The system should provide a low-cost computerised sensor-enabled mechanism to verify the contents of the parcel are verified against the expected contents.	UR-05 UR-06 UR-07

As seen from [Table 2.1](#) and [Table 2.2](#), the listed functional requirements are somehow directly or indirectly linked to the user requirements. This is purely because the system needs to perform in such a way as to ensure that the user requirements are met. Moreover, the specifications are then easily linked to the functional requirements as seen in [Table 2.2](#). The big idea behind the specifications listed in the upcoming [Table 2.3](#) is that they are merely a manner of quantifying the underlying functional requirements so that the system can be properly tested.

System Specifications

Table 2.3: BLock Box System Specifications

SP-ID	BLock Box System Specifications	FR fulfillment
SP-01	The lock system must be designed to securely store parcels of approximately 340mm x 180mm x 80mm	FR-01
SP-02	System must make use of OTP technology for access control.	FR-02
SP-03	The response time of the access control mechanism must be at least 3 seconds or less.	FR-02
SP-04	The locking mechanism must be a monostable solenoid lock.	FR-03
SP-05	The system must incorporate a high-precision weight sensor capable of detecting the presence of maximum 2kg parcels with a margin of error less than 1%.	FR-05 FR-09
SP-06	The embedded system should integrate a microcontroller with a minimum processing speed of 1.2GHz.	FR-06 FR-07 FR-08
SP-07	The embedded system should have at least 1 GB of RAM.	FR-06 FR-07 FR-08
SP-08	The IoT sensor data should be formatted in JSON and transmitted to globally accessible endpoint.	FR-08
SP-09	The BLock Box system must include a WiFi module.	FR-08
SP-10	Personal information of parties in the transaction should not be stored permanently by BLock Box and handled securely.	FR-04 FR-06
SP-11	The verification system should achieve an accuracy greater than 95%.	FR-09

The justifications of the listed specifications in [Table 2.3](#) are as follows:

Specifications Justifications

SP-01: A way to quantify the dimensions of a small package is to look at what a global company regards as their ‘small’ offering. One of biggest logistics companies in the world as this report is being written is DHL and the justification uses their standard small size [3].

SP-02: OTPs are timed passwords which satisfies the timing element of FR-02.

SP-03: User satisfaction with response time is difficult to specify without practical testing. However, it is sufficient to assume that Block Box’s processes should adhere to speeds faster than that of the load time of your typical web page as seen in the test results in [4].

SP-04: The unit must remain locked in the absence of power for security reasons.

SP-05: A weight sensor is Block Box’s mechanism for object classification. A 1% error is unlikely to change the object in question. In light of the 2kg, this is the justification as SP-01 but weight instead of size from DHL website. Although [3] list a 1.9Kg specification, 2kg is used for BLock Box to keep things simpler.

SP-06: Fernando et al. [5] proved that the Raspberry Pi 3B can be used for Blockchain transactions, therefore, its processing speed is used as a benchmark.

SP-07: Same justification as SP-07 above except the RAM of the RPi 3B is the benchmark and not the processing speed.

SP-08: Derek Ricciuto, a Co-Founder of a Chainlink oracle services company called LinkWell Nodes, comment on blockchain integration with IoT services in [6] on how to get off-chain data on-chain: ‘the easiest method would be to make your JSON available on the internet via a public URL’. This explanation is also formally available in the company documentation in [7].

SP-09: Blockchain networks are inherently online.

SP-10: Users will not interact with a system that leaks their private information.

SP-11: The verification process quantifies successful, error-free transactions which should be at least greater than the percentage of global digital fraudulent transactions which is currently 5% as mentioned in [8].

2.1.3 Acceptance Test Procedures

Acceptance Test Procedures (ATPs) are a set of test criteria that are developed to evaluate the project specification. It is, therefore, expected that each specification has at least one ATP. These are more concisely mentioned in [chapter 6](#), tested in [chapter 7](#) and evaluated in [chapter 8](#).

Chapter 3

Theory

3.1 Introduction

Ultimately the end goal of this entire investigation is to integrate blockchain technology with the well-defined world of the Internet of Things (IoT). That way the sensors that are placed on the parcel locker cannot only interpret the natural environment but the digital as well. To understand the benefits of being able to interpret the digital environment one first has to understand blockchain technology. Therefore, this chapter delves into the theoretical frameworks that define the Block Box system such as blockchain technology, decentralisation, smart contracts, oracles and their respective application in this IoT-based delivery system.

3.2 Block Box Foundational Principles

3.2.1 Asymmetric Information Economic Theory

Asymmetric information is a concept in applied economics where one party in a transaction possesses more or better information than the other party [9]. In peer-to-peer transactions especially through public ecommerce platforms asymmetric information occurs between the buyer and seller. This is because the buyer may be uncertain whether the parcel contains the promised untampered contents from the seller or not. Therefore, BLock Box system mitigates this asymmetric information by enhancing transparency and trust between the parties.

3.2.2 State space control

The domain of state space control that stems from control systems is a popular basis of this whole investigation. The states from the perspectives of the system are shown in [Figure 3.1](#).

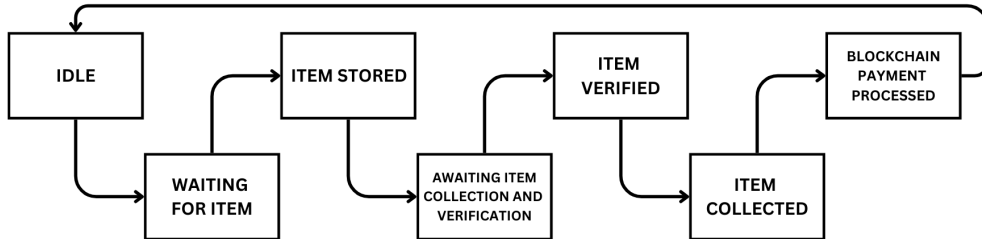


Figure 3.1: BBlock Box state diagram

The transition logic of all the states listed in [Figure 3.1](#) is based off the interactions the underlying parties have with the system as well as the inputs from the IoT sensors.

3.3 Blockchain Technology

3.3.1 Overview of Blockchain Technology

The technology was first introduced to the world as the underlying infrastructure for the well-known cryptocurrency called, Bitcoin, by Satoshi Nakamoto in a 2008 whitepaper[10]. The exact workings of blockchain are well-defined and documented, hence, this paper will not delve too deep into describing the whole technology but only the parts that are relevant to the underlying study. Rodeck and Curry describe the technology as a decentralized, distributed ledger that records transactions in a secure, transparent, and immutable manner[11]. These attributes of the technology make it an ideal solution for applications that require trust and accountability between involved parties. As is the case not only in a last-mile delivery problem but in almost every imaginable industry.

3.3.2 Blockchain Consensus Mechanisms

A consensus mechanism in the context of blockchain technology refers to the framework that the nodes of a network use when validating the correctness data. Fahim et al. [12] provide an interesting study where they detail the key differences between various consensus mechanisms. However, the primary consensus mechanisms are Proof of Work (PoW) and Proof of Stake.

Proof of Work (PoW) is the original consensus mechanism used by Bitcoin but was developed already back in 1993 [12]. In PoW, miners compete to solve complex cryptographic puzzles which require a lot of computational power and the first miner to solve the puzzle gets to add the next block to the blockchain and receive a reward[13]. This mechanism is robust and secure because tampering with a PoW network would require a tremendous amount of impractical computational power. However, the power intensity of this mechanism has led to scalability and environmental concerns.

Proof of Stake (PoS) was developed as an energy-efficient alternative to PoW[12]. Instead of relying on computational work, PoS selects validators to create new blocks based on the amount of cryptocurrency they hold and are willing to ‘stake’ as collateral [13]. Validators, therefore, have a financial incentive to act honestly because bad behavior can result in the loss of their staked funds. However, PoS may lead to a monopolisation problem as those with more staked funds would have a greater influence over the network.

3.3.3 Cryptographic hash functions

In blockchain technology cryptographic hash functions, H , play a fundamental role. These functions are essentially a mathematical algorithm that takes an arbitrary-length input data (typically referred to as the message) and outputs fixed size data called a hash value.

$$H : \{0, 1\}^* \rightarrow \{0, 1\}^n [14]$$

where:

$\{0, 1\}^*$ denotes the set of binary strings of arbitrary length.

$\{0, 1\}^n$ denotes the set of binary strings of length n .

Essentially, hashing is useful when it comes to authenticating data as it is a quicker process for your computer to check if ten documents have the exact same contents by comparing the hashes of these documents and not the sole content. The use of documents is just an example but almost all pieces of data can serve as the message and be *hashed*.

Moreover, the process is also useful for ensuring that data is not tampered with because the slightest change in the message would create an entirely different hash [15]. This unique secure inherent effect of hashing is called the ‘Avalanche Effect’ [15]. This is the very reason why this principle is used within blockchain technology especially in proof-of-work networks such as bitcoin. Another reason is that hashing is generally a one-way function unlike encryption and this implies that it is generally easy to transform a message into a hash but very difficult and computationally intensive to do the reverse operation [16].

3.3.4 Merkle Tree

A merkle tree builds on the fundamental of hashes and the term simply refers to a way of organising various hash numbers of a vast amount of transactions within a block [17]. It is a way of summing up all the transactions that occur in a block and then creating a digital fingerprint of the entire set of operation. This process works by hashing the transactions in pairs until only one hash value remains which is then referred to as the merkle root [17]. An example of this tree structure can be seen in Figure 3.2. Moreover, this summarising of transactions is particularly useful in the embedded system verification processes as the process as can be computed without having to store the contents of the entire network.

3.3.5 Data Structure

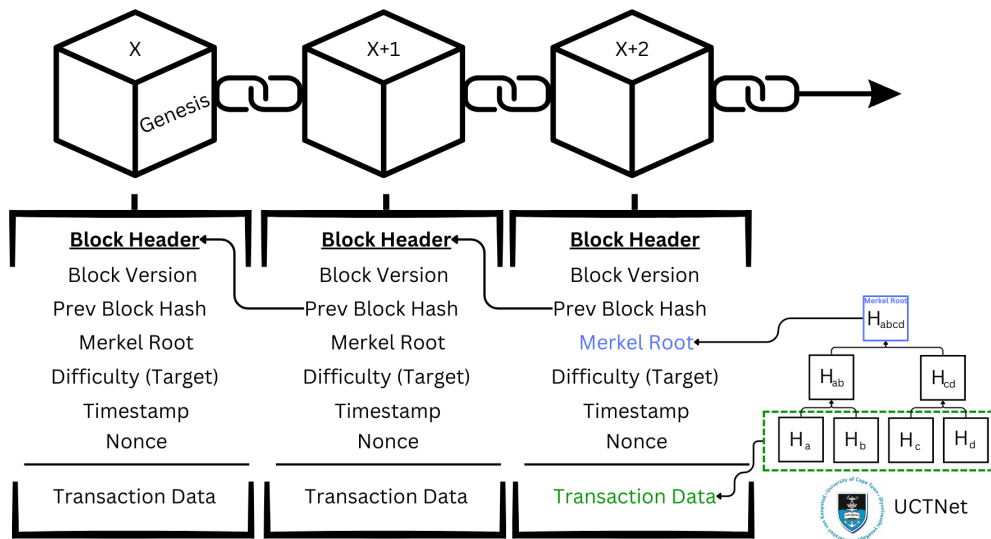


Figure 3.2: The chain structure of blocks in blockchain

As seen in Figure 3.2 a blockchain involves a train of blocks, each with a list of transactions and a reference to the previous block. As depicted this reference is in the form of the current block holding the hash of the previous block. Therefore, should anything in that block change it would alter the hash of the block which would cause all of the previous blocks to also have to be change as the tampered block would ‘break’ the chain.

Table 3.1: The elements of a block [18]

ELEMENT	DESCRIPTION
Block Header	
Block Version	Preset validation rules that the block adheres to
Previous Block Hash	Hash of the previous block's header
Merkle Root	Top (root) hash of the Merkle tree of transactions
Timestamp	Unix time when the block was created
Difficulty	Measure of how difficult it is to mine a block in proof-of-work
Nonce	Number that is used once and assigned to a block during mining
Block Body	
Transactions	Validated transactions included in the block

Table 3.1 provides a Hi-level description of the elements shown in Figure 3.2.

3.3.6 Decentralisation and Distributed Ledger Technology (DLT)

The decentralisation nature of blockchain implies an absence of a central controlling authority. This attribute is particularly beneficial in applications involving IoT devices and physical assets, where decentralisation can prevent centralised control or data breaches from compromising the entire network.

Whereas the distributed part of the technology refers to the widespread peer-to-peer network where each participant maintains a copy of the ledger. This ensures that all nodes see all the underlying transactional activities which enhances trust within the network. The only way any changes can be made to the ledger is if an entity gets control of 51% of the network. This is highly unlikely as in a PoW network as the attacker would need 51% of all the computational power which becomes nearly impossible in a wide scaled network. In a PoS case 51% of the tokens would need to be staked and mined which is a process that becomes extremely expensive the more liquid the network gets.

3.3.7 Smart Contracts

Smart contracts are parametric contracts between two parties that self-execute based on meeting preset conditions. Therefore, these are crucial for the underlying study of integrating blockchain technology with IoT devices on a parcel locker. This is because a smart contract can automatically trigger an action, such as unlocking the secure locker or transferring funds for the sale of an asset when IoT sensors have automatically verified the asset for the buyer as the item matches specific conditions.

3.3.8 Oracles and Data Feeds

While blockchain technology is highly secure and reliable for managing digital assets and contracts, it inherently lacks access to external data, which is crucial for interacting with the physical world. There has to be a mechanism constantly feeding the blockchain with the IoT sensor data. This bridge between the blockchain and off-chain resources is addressed by oracles which according to Coinbase are known as third-party services that provide smart contracts with external information [19].

Chainlink is a popular network that is known to offer decentralized oracles that connect smart contracts with real-world data. The workings of this network is in such a way that node operators are the ones to provide this off-chain data to consumers and Chainlink would then aggregate similar data from the providers in order to enhance the integrity of the data. This revolutionary concept is made possible through the use of external adapters within Chainlink. Essentially, external adapters are modular elements that can be embedded into a Chainlink node to fulfil a practical purpose [20]. The Chainlink article on [20] describe external adapters as an open-source package but for Chainlink nodes. Therefore, users are able to take customisable elements and embedded them in a Chainlink node using a *job specs* as defined in the Chainlink docs [21]. These *job specs* can be set up to fetch requested off-chain data from specified authenticated and secure APIs.

Therefore, in light of Block Box, the system takes the approach of developing a secure public web-server endpoint with the system state data in JSON format to which a job spec could be developed to be able to use the data in the transaction-based smart contract between the buyer and the seller [6]. Although block box does not aim to create its own Chainlink job spec the process is simulated to ensure that a fully functional last-mile delivery system can still be developed.

Chainlink also offers public oracles predominately in the exchange rate domain under the umbrella called price feeds. Block box makes use of this to get USD/ETH. A normal HTTP GET request from a centralised site was used to get ZAR/USD as this is a South African based investigation and Chainlink does not have a public oracle for ZAR/ETH yet but this can then be found with the following formula:

$$\frac{\text{ZAR}}{\text{ETH}} = \frac{\text{ZAR}}{\text{USD}} \times \frac{\text{USD}}{\text{ETH}}$$

Chapter 4

Literature Review

4.1 Secure Parcel Locker Systems

At the time this review is compiled DHL is the leading logistics company with a market capitalisation of over \$50 billion dollars [22]. According to the company's website they define parcel lockers simply as secure, automated storage units in centralized locations [23]. They are the perfect example of how they have leveraged the parcel locker to solve several last mile delivery problems such as costs incurred in multiple locations deliveries, missed deliveries, stolen packages, etc [23]. However, although it may seem trivial it is imperative to have a look into the how these lockers have traditionally been used before looking at modern approaches such as those adopted by DHL.

4.1.1 Traditional Parcel Locker Systems

Traditional parcel locker systems have been around for decades and before they were used as a delivery solution they were used as a common form of secure storage even found in a Victorian Era wreckage of a British warship back in 1848 [24]. These are typically made of industry graded steel [25] and are found at post offices, apartment buildings, gym and retail stores. Users would be able to store their items or collect their parcel using a unique key assigned to them. Srinivasan et al. discuss how in most banks customers can access their locker only with the physical assistance of an employee as the locks are still manually operated. [26] These systems rely heavily on manual processes and physical security measures, which can be time-consuming and less efficient. This reliance on simple access codes for locker access lessens the robustness of the security of the overall system.

4.1.2 Modern Parcel Locker Systems

Modern parcel locker systems have evolved significantly because of a wide range of technological advancements. These systems are similar to the ones adopted by logistics companies such as [DHL](#), [Amazon](#), [DSV](#), [pudo](#) and many others. These systems are essentially a combination of electronically controlled locks, IoT devices, robust enclosures and user-friendly software interfaces that not only make the system convenient for the user but also enhance the overall security.

Moreover, [section 4.6](#) elaborates on how the technology within the blockchain realm that is capable of taking these modern lockers and enhancing the efficiency of the whole system by removing the trusted third party between buyers and sellers.

4.2 Locking Mechanisms

The level of security a parcel locker has is highly dependant on the type of locking mechanism that is being used. Traditionally, these lockers have used simple mechanical locks, however, technological advancements has led to more secure electromechanical locks. These locks are a lot more efficient and sophisticated as they can be controlled remotely or via a user-enhanced input such as passwords, OTPs, or biometric data as described in [subsection 4.3.1](#).

4.2.1 Mechanical Locks

Mechanical locks are the traditional locking mechanisms that encompass physical and mechanical components such as key-locks and padlocks. These have long been the main physical security access control mechanism for parcel lockers. Although these are still widely in use mainly because of their simplicity and cost-effectiveness they are a security risk, especially in cases where tampering is a concern [27]. These security risks range from key-loss, lock-picking as well as simply being able to use a bolt-cutter for padlocked boxes [28]. This rationalisation is further heightened by Miehl [29] in stating that door locks, which happen to be mechanical locks, are susceptible to physical manipulation. Therefore, they would not only be undesirable but unsuitable for sensitive and valuable systems such as parcel delivery lockers.

4.2.2 Electromechanical Locks

In response to the security vulnerabilities of traditional locks, modern locking mechanisms have been developed and widely implemented by society. These include magnetic and electric strike locks. The former is commonly known as fail-safe where as the latter is called a fail secure lock [29]. Moreover, magnetic locks are commonly known as fail-safe locks which means they need power to lock a door and unlock when they is no power [29]. Electric strike locks on the other hand usually need power to unlock the door as they rely on a solenoid to keep the door locked. The latter falls under the solenoid lock family and the former is commonly referred to as a maglock [29].

4.3 Access Control Methodologies

An access control mechanism for a locker system comprises of a combination of advanced technologies and a secure locking mechanism as discussed in the previous chapter. The authentication of a user can often be separated into three categories [30]:

- Something they have - Possession-based Authentication
- Something they know - Knowledge-based Authentication
- Something they are - Biometric-based Authentication

4.3.1 Possession-based Authentication

Possession-based authentication are the access control methods that require the user to hold a specific item and than that item is the one interacting with the system to grant the user access. This can be as simple as a key or as complex as an NFC/RFID enabled chip as in [31, 4]. NFC and RFID are useful because are a quick contactless technology where an authorised card is tapped against the locker's reader for access to a parcel locker [31, 4]. QR codes as shown by Tahyuddin et al. [32] are also another form of possession-based authentication for parcel lockers as they are normally used with an interface where the user scans the code and completes instructions to open the

locker such as sending an access request to an authority as in the study. Therefore, the user does not only need to have the code but they do need to possess a reader such as a smartphone to interact with the system.

4.3.2 Knowledge-based Authentication

Knowledge-based authentication methodologies are those that rely on information that only the user should know. Therefore, a non-exhaustive list of these methodologies include PIN codes, OTPs, passwords and security questions. The description and operation of these methods are well-defined and well spread in society because of their simplicity. However, despite their global adoption, they house a lot of underlying vulnerabilities.

An example of a drawback to knowledge-based authentication is the huge risk in information leakage. Information leakage is essentially when an unauthorised party gets access to the pin or password [33]. Logically, an intruder can get access to a locker if a user has written their password somewhere on a piece of paper or a storage medium that happens to get hacked. Moreover, they are even methods that are used to get the password without the user causing the private information to leak via human error. An example of this is highlighted in a study conducted by Balagani et al. [34], where they showcase that even masked characters displayed during an entry of a pin or password can be used to crack the underlying password. The PILOT attack platform that they developed was able to recover passwords and PINs from taking video recordings of the inputted passwords and then doing an inter-keystroke timing analysis on the video [34]. Parcel lockers normally relay on knowledge-based authentication methods such as passwords and if an attacker can get footage of the locker's interface they could essentially use a technique similar to that of a PILOT attack to gain access to the contents of the locker.

However, Mostakim et al. [35] and [36] propose a one-time-password (OTP) based parcel locker as a knowledge-based authentication method that is used to counteract the possibility for information leakage to occur. An OTP is a dynamic password that always changes and a unique one is sent to the user each time they try and access a system [35]. In a parcel locker scenario this would indeed be immune to a PILOT [34] attack but adds a layer of inconvenience to the user. This is because they would consistently have to be interacting with the external system that provides them with the OTP every time they want access the contents of their locker. This setup is not ideal where a locker is used for personal use purposes and the same user has to access the same locker multiple times. However, OTPs work well in lockers used for delivery services as parties need temporary access.

4.3.3 Biometric-based Authentication

The OTP locker system mentioned in subsection 4.3.2 by Mostakim et al. [35] also utilised face detection technology which is a biometric-based authentication methodology. The most common form of biometric-based authentication is fingerprint technology which is applied within the parcel locker domain by Narayanasamy et al. [37] as a means for customers to confirm the collection of their respective parcels.

4.3.4 Multifactor Authentication

Multifactor Authentication is a common security process of where the user is required to provide two or more independent forms of verification before they can access a system. Contrastly, Sobur et al. [31] discuss a single factor authentication approach for a locker by using RFID technology. This locker was to be used for securing personal belongings in public places such as gyms, malls and offices. In a nutshell the system also had a keypad for admin control and sensors to detect when a locker is in use [31]. However, in a place that needs more security than a public area such as a bank a MFA process needs to be used. An MFA approach to Sobur's [31] solution is

proposed by S. Sridharan et al. [26] as they investigate a secure locker system also using RFID with the addition of password and GSM technology for real-time user verification and authentication. Therefore, it can be seen that the type of environment and the contents being secured determines the level of security to be applied when using designing the access control mechanism. This reasoning is heightened even further by Detroja et al. [38] and Htwe et al [39] as they went on to even add a biometric-based authentication layer to the security issue for their bank lockers by using fingerprint technology. Parcel lockers may not need the extreme security provided by lockers in banks but they need to be a lot more robust to intruders than an office or gym locker.

4.4 Applications and Use Cases for Parcel Lockers

Parcel lockers are used in a vast number of applications ranging from e-commerce to public and private use.

4.4.1 E-commerce Utilisation for Last-Mile Delivery

Home deliveries are currently the most commonly used form of last-mile delivery service for the e-commerce industry, however, Iannaccone et al. [40] prove in their study that the tides have been shifting towards parcel lockers for the new generation consumer. This is because they showcased how parcel lockers grant consumers greater flexibility on when and where they want to collect their package. This is utterly important in urban areas as flexibility is seen as a commodity. However, urban environments are not the only places that benefit from parcel lockers as rural area consumers also have a lot to gain in this the adoption of parcel lockers as a last-mile delivery option [41]. This is because in these areas logistic service delivery is inadequate as roads and address systems are poor or non-existent, therefore, making the parcel locker an ideal solution [41].

Furthermore, to heighten the use cases of these lockers, Peppel et al. [42] use a mathematical based simulation to show that in regions of more than 20 000 people the parcel lockers save 8.7% more costs and lead to a 5% less emission of CO₂. Therefore, the use of parcel lockers are not only good for consumers but also a means to make the logistics industry a lot more sustainable. Moreover, Van Duin et al. [43] discuss another advantage of these lockers within the e-commerce industry on how they alleviate the common issue of missed deliveries when the recipient might not have been available or not at home.

Nevertheless, apart from recipients not being at home there is also a lot of crime that occurs in the stealing of packages in home deliveries [44]. Drew McKay [44] illustrates how legislation is used to stop porch pirates from stealing packages for a not-at-home consumer scenario problem. However, legislation alone is not enough as it has not stopped porch pirates from going on with their malicious activities. Legislation with the help of parcel lockers is the ideal solution to stop porch pirates as discussed by Sang-An et al.[45] as they illustrate how “online shoppers can prevent package loss by using a secure locker”. The fundamental basis of the study proves how parcel lockers drastically decreases the chances of package theft.

4.4.2 Personal Use within Public Spaces

Ecommerce is not the only use-case for secure parcel lockers as already discussed in the RFID-enabled prototype Sobur et al. [31] created for public use in public areas such as shopping mall, religious institutions, gyms, postage services, etc.

4.4.3 Private Locker Boxes

Secure parcel lockers or also referred to as locker boxes are being increasingly popular for private use at homes. Whether one is a lawyer who needs to store away secure documentation or an individual who needs to store away cash, jewellery or any other valuable asset. Furthermore, a combination of the private use of the lockers with the ecommerce use is a trendy application of customers having a parcel locker outside their residence to collect deliveries for them in their absence [46].

4.5 Parcel Verification Methodologies

Customers tend to use a courier service with an expected delivery deadline as according to a DHL-Deloitte [47] audited global trends report 48% of customers abandon their cart if the site has no satisfactory delivery option. Therefore, it is of utmost importance that the customer can fully trust that their ordered item will be in their hands in the indicated time frame. Moreover, besides having to meet delivery times, logistic companies have evolved to updating the user on the several verification steps their parcel has undergone in real-time using a unique tracking number [23]. Therefore, customer trust and satisfaction is of utmost importance within the supply chain and logistics industry which is yet again seen with 77% of customers wanting to know who their delivery provider is [47]. This section explores the key verification methodologies used with a focus on IoT sensor technologies that play a critical role in the verification of the contents of a customer's parcel.

4.5.1 Types of Sensors for Content Verification

Weight sensors:

The use of weight as a parcel verification mechanism is not a newly found idea in the logistics industry and an application is illustrated by [36]. The concept works by comparing the listed weight of a package and ensuring that the weight is consistent from the initial stages of the order to just before it lands in the hands of the end user. Therefore, any significant change in the weight from when the seller packaged the product to the item being in the last-mile process could indicate potential product tampering. Typically, these sensors are placed in the parcel lockers themselves in order to continuously monitor the parcel's weight.

Cameras (Computer Vision)

Computer vision technology has enhanced rapidly and continues to see tremendous breakthroughs. The technology is all about using cameras and optics to grant a computer the ability to be able to detect the natural environment. Hence, in a parcel locker scenario, this can be used to verify whether the contents that are listed on the seller's description are indeed the ones in the hands of the buyer. An image of these contents can be classified using and of the image classification models mentioned by [48].

NFC (Near Field Communication):

NFC technology is very well documented and utilised in modern society. It was briefly mentioned in [subsection 4.3.1](#) as an access control mechanism but in the proximity of parcel verification, these NFC tags can be placed on the package or even on individual items within the package [49]. Furthermore, Smiley [49] also touches on the use of RFID for Tamper-Evident Packaging which would be a useful technological implementation within parcel delivery systems to ensure that the items the buyer receives have not been tampered with.

4.6 Blockchain Technology and Smart Contracts

The integration of blockchain technology and smart contracts for last-mile delivery is a revolutionary worthy task. This is because blockchain offers a decentralised, unchangeable digital ledger that can increase transparency and security by recording every transaction and state of the goods in question [11]. Smart contracts on the other hand can automate payment releases and access control procedures through its coded self-execution philosophy.

4.6.1 Need for Blockchain and Smart Contract Integration

The increasing demand for secure, transparent and efficient transportation methods is the underlying need to integrate blockchain and smart contracts within logistics services. The tamper-proof characteristic of the blockchain technology that can address issues such as fraud and lack of transparency. Manual interventions such as payment processing, escrow services, access control and storage of verification content could be automated with the likes of smart contracts. [50] investigates what this integration would mean to supply chain companies to be able to improve their efficiency, increase trust and possibly be able to increase profits due to reduced costs.

4.6.2 Integration with Physical Systems

Blockchain technology is fundamentally a digital technology so it is difficult to integrate it with real-world physical systems as mentioned in subsection 3.3.8. This incorporation of physical systems is, however, needed in order to ensure that full transaction transparency in all areas of the last-mile delivery process is achieved. The challenge lies in ensuring that the physical actions such as the deployment and shipping of goods as well as their verification is accurately recorded on the blockchain. Therefore, for this to be feasible an interface is needed to connect the digital blockchain world with the physical world. This implies a connection between IoT devices and the blockchain is needed and this can be found through data feeds using oracle technology.

Moreover, an example of this integration is provided by Grecuccio et al. [51] in a study where they propose a software framework that allows IoT devices to be able to communicate directly with an ethereum-based blockchain. The study [51] mainly focuses on the supply chain of the food industry and documenting the whole process from production to delivery on the blockchain to enhance trust and visibility using a web-server and a web3 ethereum API. This enhances the traceability of perishable good which in turn not only reduces the risk of fraud but also improves the overall safety of the underlying food [51]. Casado-Vara et al. [52] and Baralla et al. [53] are similar food supply chain studies but focusing on the agricultural industry. Last-mile delivery peer-to-peer logistics services can learn a lot from the food industry in this context as traceability is a key factor from both lenses.

Logistic process automation using smart contracts

Therefore, in the context of supply chain industry, a similar approach can be taken to ensure that the integrity of the parcel contents is maintained throughout the delivery process. This can be seen in an article presented by [54] which showcases how the retail giant Walmart filed a patent for a blockchain application used to track its various shipments. This application includes a device to store the contents of shipment similar to [55] as well as the full state of the parcel with key addresses such as the private key of included stakeholders from clients to couriers are hashed securely onto the block. However, since BLock Box is only concerned with the last-mile delivery process, Cekerevac et al. [56] discuss this complexity and highlight the existing correlation between the quality of the delivery service with the integration of blockchain technology. This correlation in [56] stems from the technology's trait in enhancing the traceability of the supply chain.

This traceability is made possible through the mentioned smart contracts in [subsection 3.3.7](#). Hasan et al. [55] evaluate a broad but efficient supply chain management system that utilises smart contract technology to manage the conditions of shipments, automate payments and issue refunds in cases where certain conditions are not met. The self-executing process that runs on the blockchain have the agreement terms of the buyer and seller directly embedded within its underlying code. Therefore, when these parametric conditions such as the parcel being a specific weight or the camera detecting specified items are met the contract executes. This execution as mentioned can be a payment release to a seller provided that conditions set by the involved parties are met.

4.7 The future of last-mile delivery

The Block Box system developed within this report integrates IoT devices with blockchain technology as a solution to the last mile delivery asymmetric information problem¹. Although suggested improvements to the BLock Box system is listed in [chapter 10](#) the last-mile delivery industry is also on a non-ending exponential growth curve.

4.7.1 Last-Mile Delivery Innovations

Last-mile delivery as discussed is globally recognised as one of the most challenging and costly segments of the supply chain cycle [57]. Therefore, this leads to the natural development of several innovations as mentioned by [57] to try and address these challenges and make last-mile delivery more efficient.

Utilisation of advanced robotics for parcel handling

Boston Dynamics in their developments of [Atlas](#) and Tesla with their bot [Optimus](#) are illustrating revolutionary advancements in humanoid robotics which could be applied for last-mile delivery processes. Agility Robotics has already testing humanoids in Amazon warehouses [58] before they can be implemented for the last-mile.

Data analytics and AI

Data analytics and AI are playing an increasingly crucial role in the optimisation of the last-mile delivery process [59, 60]. This is because we live in a world where information is the new gold. However, just like gold has to be mined so does data has to be analysed to be able to attain interpretable, useful information. AI used as a tool for data analytics allows you to harness computational power to extract valuable information from a large amount of data. Chu et al. [59] discuss these optimisations that AI-solutions can bring to the last-mile, particularly in the domain of delivery route optimisation.

¹[chapter 3](#)

Chapter 5

System Design

5.1 System Hi-level construction

As shown in the sequence diagram in [Figure 5.1](#) the BLock Box system is a secure, automated solution designed for product exchanges between sellers and buyers using IoT sensors and smart contract technology. The seller places the parcel in a locker where a weight sensor detects its presence and notifies the buyer for collection. All system states are stored along with the timestamp of when a particular event occurs as can be seen in [Figure 5.1](#). The buyer would then require the correct authentication to access the locker. Upon successful authentication, the buyer takes their parcel and the system updates its state to ensure that this collection is noted. A smart contract then releases funds to the seller based on the condition that the item has successfully been collected by the buyer.

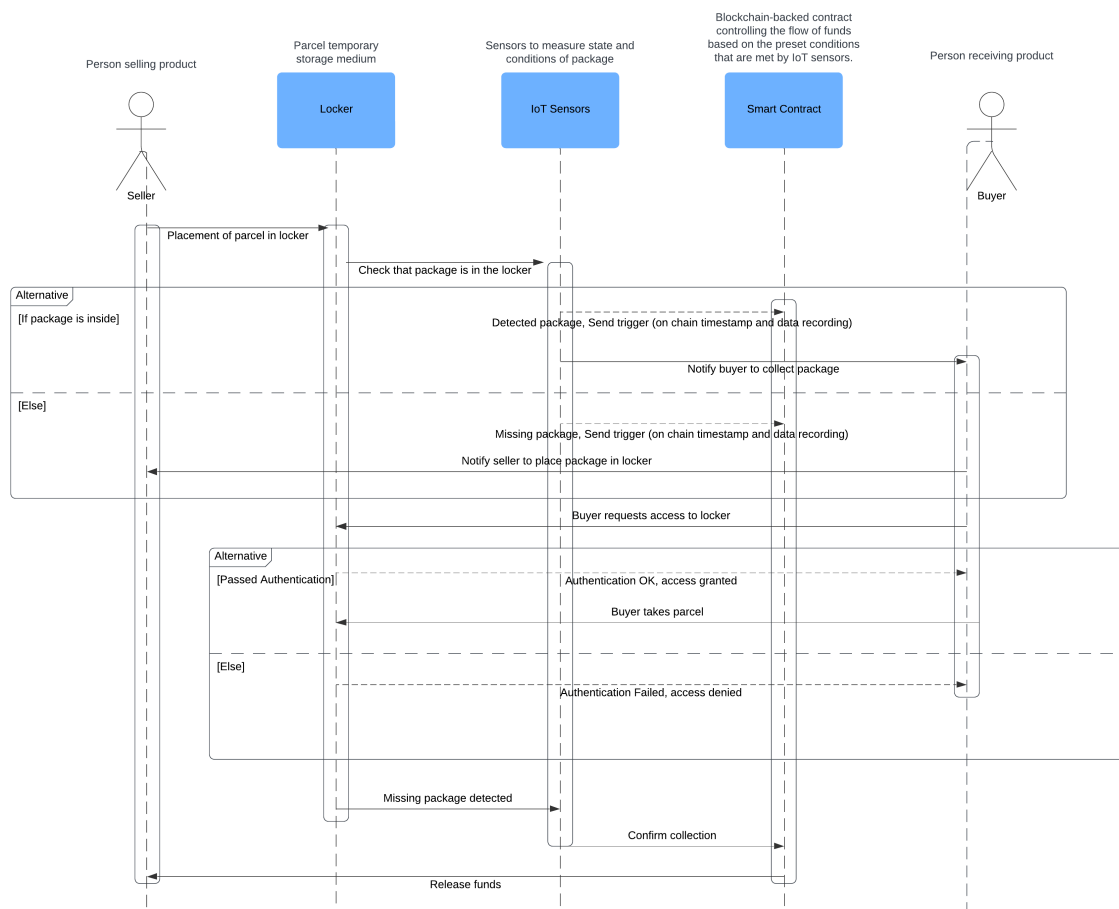


Figure 5.1: Sequence diagram for BLock Box

This process of the interaction between the buyer and the seller that BLock Box establishes can also be represented in a simple flow diagram as shown in Figure 5.2.

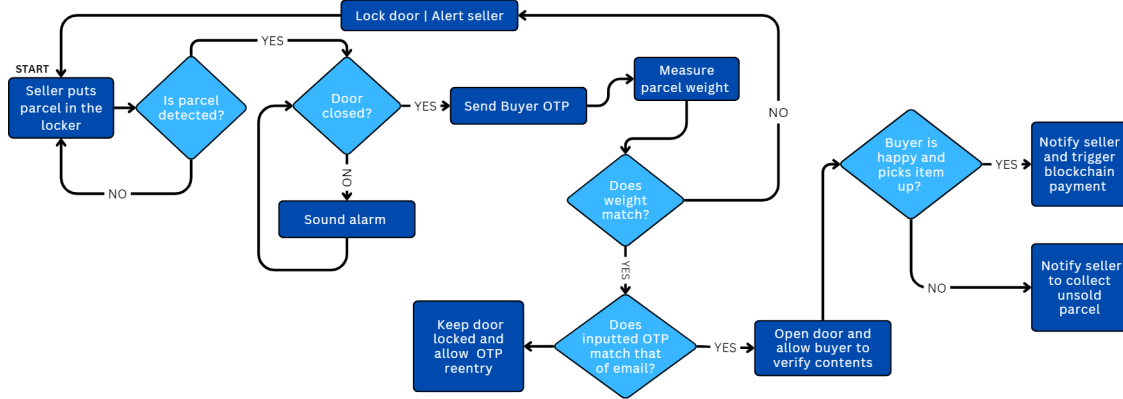


Figure 5.2: Flow diagram showing buyer and seller interaction

5.2 System Sub-module Breakdown

The Block Box system can be split into four different sub-modules. These are **power**, **process and control**, **User Interface** as well as the **enclosure** sub-module.

The **process and control** sub-module is responsible for managing the various interactions between the different hardware components of the Block Box system. Therefore, this sub-module ensures that all hardware components work seamlessly together to ensure that efficient and secure access control of the locker is feasible. The software operation of the parcel verification system is also a part of this sub-module.

The **User-Interface** sub-module is heavily linked with the process and control sub-module, however, it served more as a vessel that allowed the system communicate with its external environment. In other words, in cases where this sub-module was responsible for gathering OTPs the process and control sub-module was then responsible for the validation of these OTPs. Therefore, the buyer and seller physically interacts with Block Box through this sub-module but the actual computation of the interaction occurs in the process and control sub-module.

The **power sub-module** is responsible for ensuring that all components of the Block Box system receive their appropriate rated operational voltage and current. Moreover, aside from the adequate distribution of power this sub-module ensured that all the voltage is regulated appropriately.

The **enclosure sub-module** encompasses the physical locker that houses all the underlying components of the Block Box system. This system provides the physical security and durability of the prototype of the locker. Although the production-version of BLock Box is discussed in chapter 10, a less secure, budget-friendly prototype is developed in this investigation as the focus is more on solving the asymmetric information payment problem.

These four sub-modules work together in order to ensure that the Block Box system is coherent and efficient. The interconnections of these sub-modules can be seen in [Figure 5.3](#).

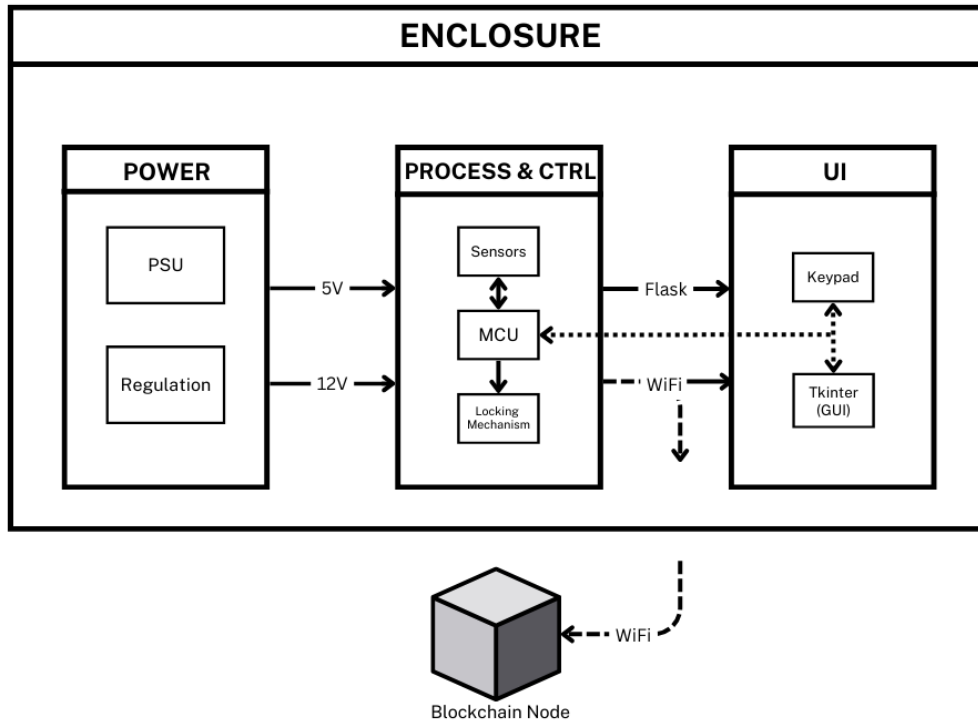


Figure 5.3: High-Level Block Diagram of the Block Box System

Figure 5.3 is a high-level block diagram of the Block Box system. This diagram shows the flow of data and signals between the three different sub-modules.

5.3 Design Choices

It is imperative to mention that concise design choices had to be made for each sub-module. These choices are made to coincide with the system specifications as listed in [chapter 2](#). A combination of these design choices also have to ensure that the overall R2000 budget of the project is not exceeded. This section takes the approach of separating the different design choices in their different sub-modules as shown in [Figure 5.4](#).

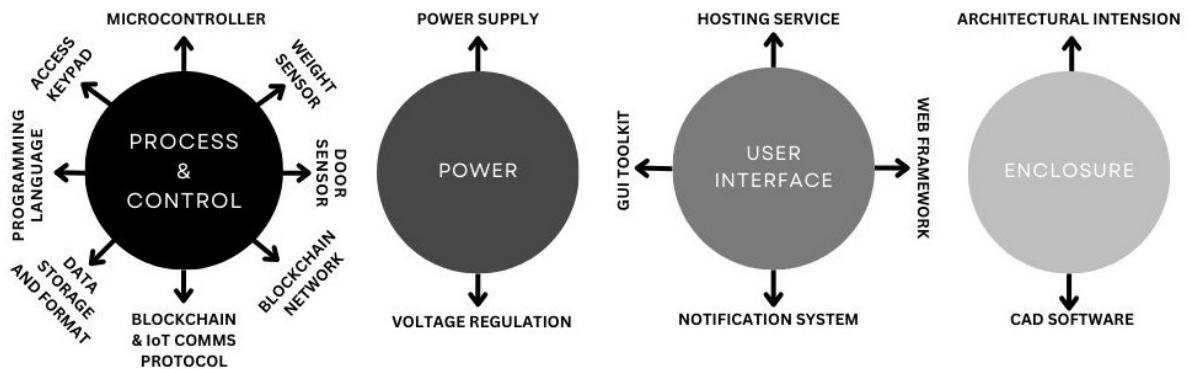


Figure 5.4: Sub-module design choice overview

5.3.1 Process And Control Sub-Module

Micro-controller selection

The Raspberry Pi (RPi) 4B 1GB is the micro-controller of choice for this project. This is simply because the focus of the project is not on working with only low-power electronics as a practical parcel locker system would be connected to the main power supply. Moreover, a system specification is to be internet-compatible and although this is achievable by using the comparable ESP32-Wroom-32, Arduino Portenta, Beaglebone Black or any other micro-controller with a WiFi module, the benefit of using the Pi 4B is its balance between cost, processing power and memory. These aspects are particularly important when it comes to the blockchain integration. Moreover, a short evaluation based off of these performance metrics of the mentioned micro-controllers is shown in [Table 5.1](#).

Table 5.1: MCU comparison table

Microcontroller	Processor Speed	Memory	Internet-Compatible	Cost
Raspberry Pi 4B	1.5GHz	1GB+	Yes	Moderate
Arduino Portenta H7	480MHz	8 MB	Yes	High
ESP32-Wroom-32	240 MHz	520 KB	Yes	Low
Beaglebone Black	1GHz	512 MB	Yes	High

As can be seen in [Table 5.1](#), the Beaglebone Black was the nearest competitor to the Raspberry Pi but lost out in being selected because of its slightly lower processing speed whilst having a higher cost. The extensive documentation and community support that the Raspberry Pi provides was also a factor in considering it as BLock Box's micro-controller. It is worth mentioning that the Raspberry Pi is generally regarded as a microprocessor and not a micro-controller because of its mini-computer configuration, however, for the sake of this report the term micro-controller is consistently used.

Programming Language

Python was selected as the programming language of choice for the BLock Box. Although, the syntax of the language is indeed simple its real power is derived from the wide range of libraries that are developed for it. One such library that played a role in choosing python as the language of choice is the Web3 library made for blockchain integration.

Weight sensor selection

As part of selecting a weight sensor the parallel beam load cell was compared to a multi-point platform (four load cell) setup which are both showcased in [Figure 5.5](#) and [Figure 5.6](#) respectively.

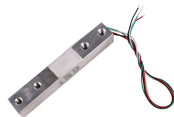


Figure 5.5: Parallel Beam Load cell [1]

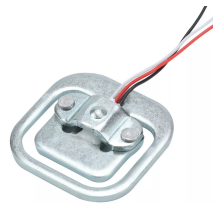


Figure 5.6: Platform Load cell [2]

Although the parallel beam was used for testing of the system because of its simple wiring and only having to put one cell in the center of the platform, the multi-point platform load cells were the ones selected weight sensor design for the BLock Box system. This is because the weighing bed within BLock Box is fairly large and the multi point platform load cell configuration is theoretically able to slightly compensate for uneven loading. The accuracy and resolution of the two load cell types were similar at $1.0\text{MV/V} \pm 0.15\text{MV/V}$. Moreover, the platform configuration was a lot more suited for a flat, scale platform than a parallel beam.

However, it is imperative to mention that the platform load-cells that were available in market were 50 Kg load cells which was not ideal. This load cell was used but since Block Box is rated only for 2Kg max weight, having a high maximum threshold is a waste and may influence the accuracy. This threshold further increases as the multipoint configuration platform load cells are connected in a wheatstone bridge configuration so the max weight becomes $50\text{Kg} \times 4 = 200\text{Kg}$. Therefore, although the platform load cell is the type of load cell recommended for BLock Box as mentioned, the specific one that was bought for this investigation is not ideal. The HX711 was chosen as the amplifier of choice for these load cells because of its immediate availability.

Door sensor selection

A magnetic reed sensor was selected as the door sensor for the BLock Box system. An alternative that was considered is an ultrasonic sensor, however, this was not selected because an ultrasonic sensor could potentially trigger a ‘closed door’ case from a user simply standing in front of the parcel. Whereas a magnetic reed sensor is more accurate in the sense that the sensor would only trigger when the magnet that is mounted on the door touches the strike plate where the sensor is located.

Access Keypad selection

The 4x4 tactile matrix keypad in this Block Box system was used as the access control mechanism which the buyer used to enter an OTP to unlock the locker for collection purposes. It is worth nothing that since Block Box uses OTP technology, knowledge-based authentication system as mentioned in 4, other access control types were not considered.

Nevertheless, the 4x4 version was selected over the 3x3 counterpart to have more key options to enhance the security of the OTP for the end-user. Furthermore, a tactile keypad was selected over the cheaper membrane version to provide the end-user with a physical response when a key has been inputted to which aids in decreasing the probability of entering the incorrect OTP. A touch screen service with a display to showcase the inputted OTP would have been the ideal solution to enhance the user experience but it was not considered because the single touch screen module for the Raspberry Pi currently goes for approximately [R1 650.00](#) which would be the entire budget.

Blockchain network selection

Ethereum was selected as Block Box’s layer 1 blockchain network of choice with the testnet Ethereum Sepolia being the actual implemented network. Layer 1 simply refers to the base blockchain network. Layer 2 platforms would be those like polygon which are built on a layer 1 protocol but solve a particular Layer 1 issue like the gas fees of Ethereum.

Ethereum was chosen as the network is extensively documented with a very active community base. Moreover, the network offers smart contract support which is a crucial basis for its selection. Bitcoin was not considered as a platform because the network's scripting language was not designed for smart contract integration. Moreover, Solana was considered as a network as it has way faster transaction speeds than Ethereum. However, as good as Solana seems to in comparison to Ethereum, the network is still fairly new and so community support is not as vast as with Ethereum.

Communication Protocols for Blockchain integration of IoT data

HTTP and MQTT were the two communication protocols that were considered.

Table 5.2: HTTP and MQTT comparison table

Protocol	HTTP	MQTT
Message Size	Large	Small
Connection Type	One-time	Always-on
Power Usage	High	Low
Speed	Slower	Faster
Best Use Case	Occasional, large state updates	Small, frequent, real-time state updates

As seen in [Table 5.2](#) it the faster, lightweight approach of the publish/subscribe MQTT protocol justifies why it is a popular IoT protocol. Although the massive downside of this protocol is that it still leverages a centralised message brokerage service which impacts blockchain's ultimate goal of true decentralisation. Http which works on the request/response service on the other hand is also physically centralised in that there is a central server that is responsible for initiating the communication service. However, the technology is a lot simpler to implement and this simplicity was the main basis for its selection.

System Data formatting and storage

The data that is made available through an endpoint using the HTTP communication protocol as mentioned in [section 5.3.1](#) but this data still had to be formatted in a specific way before the data can be pulled by an API to be able to be integrated within some smart contract. JSON was used as the format in which the sensor data and the various states of the BLock Box system would be saved as not only because it is a highly human-readable and easy to parse but also because of its widespread adoption which makes it a universally acceptable data format.

A detailed examination of internal, external flash memory and SD cards was done to find the best on-edge storage media for the sensor data. The most inexpensive solution was found to be SD cards. In light of internal flash memory, the RPi 4B+ does not have any so this is not an option. However, even if the flash memory was available on-board it still would not be ideal because it usually has a low, limited capacity, therefore, not recommended for storing large amounts of sensor data especially image data should the Block Box system be adapted to computer vision verification methodologies in the future. Moreover, external flash memory in the form of USB or SSD was considered only as a backup storage medium to which USB technology was chosen because of its comparative inexpensiveness.

5.3.2 Power

Power supply selection

In order to determine the power supply to be implemented in the system a power budget has to be computed to be able to see the net power requirement of the system. Therefore, [Table 5.3](#) outlines BLock Box's power requirements on a per component basis.

As illustrated in the power budget in [Table 5.3](#), the solenoid lock is the component with the highest voltage of 12V. Therefore, the chosen supply has to be a 12V power supply unit. The net current that the components need is 3.6415A, therefore, a 12V, 4A power supply unit (psu) would be adequate to power the BLock Box system. However, the psu that was used in this investigation is a 12V, 8.5A psu because of its immediate availability which is rated at a higher current but will have the same effect because components only pull the amount of current they need.

An advantage of using a psu over a battery is that a psu already comes with protection circuitry with a stable voltage output, therefore, there is no need to design extra supply regulation circuitry. Moreover, a backup power supply for BLock Box is not needed as the system auto locks when system fails which is enough until technician comes.

Table 5.3: Block Box Power Budget

Component	Description	Voltage (V)	Current (A)	Power (W)
Raspberry Pi 4B	Microcontroller	5	3.0	15.0
Solenoid Lock	Locking mechanism	12	0.6	7.2
HX711 Amplifier	Load cell interface	5	0.0015	0.0075
Keypad	User input device	5	0.030	0.15
Piezo Buzzer	Indicator	3.3	0.01	0.033
Total				22.39

Voltage regulation

Although the psu takes care of taking the grid AC voltage and providing a stable 12V DC voltage, the Raspberry Pi still needs a stable 5V, 3A supply. Various buck converters and voltage regulators were considered. The Hobbywing 5V 3A UBEC step-down converter was selected as the regulator of choice. A simple 5V buck converter alone with no current limiting circuitry is not sufficient because the power is a constant so the converter would replace the 12V with 5V but increase the current proportionally damaging the RPi.

5.3.3 User Interface

GUI Toolkit

BLock Box uses the Python library Tkinter to provide a GUI for the buyer and the seller to interact with each other and the system. Although PyQt is synonymous with commercially-built applications and is much more powerful than Tkinter it does have a steeper learning curve. Hence, the simplicity of Tkinter is the basis of the decision to choose it as BLock Box's GUI Toolkit.

Web Framework

Flask was chosen as the web framework for Block Box due to its lightweight nature and ease of integration. Martin [61] compares this framework with Django and Pyramid and a foundation that is formed is that Django is good for large-scale projects and Pyramid is good for small or large but adds a layer of complexity.

Hosting Service

Ngrok was selected as the hosting service as the platform provides HTTPS tunnels, which ensures that the transmitted data is encrypted and secure. This is particularly useful during development and testing phases, as is the case because BLock Box is a concept project and not a production-ready one.

Alternative hosting services like Azure or Amazon Web Services (AWS) could be used to deploy the application to a production environment. However, these services require more complex configurations and may often have additional costs.

Notification System

Telegram bots were integrated into Block Box as the notification system because of their simplicity, widespread adoption and robust API. The platform is free and capable of providing real-time user system notifications. SMS was also looked into, however, a GSM module would need to be purchased and GSM modules for RPis are quite expensive. Email on the other hand often has the caveat of increased latency, therefore, leaving the Telegram API as the ideal option.

5.3.4 Enclosure

The enclosure sub-module concerns itself with the physical housing of the project. BLock Box is a last-mile delivery locker box system, therefore, the enclosure should have a level of tampering resistance to it. Therefore, factors such as a durable, sturdy material selection with concealed hinges are just the basic considerations.

Architectural Intention

The idea behind the housing of Block Box was to combine elements of a normal safety box with a typical drawer with a platform. This drawer system was established to give the buyer a platform to be able to verify their parcel without the need of picking the parcel up from the platform which would trigger the blockchain transaction.

CAD software selection

Fusion360 by AutoDesk was used as the CAD modelling software. Although, the task at hand composes of just modelling a simple box, therefore, any other modelling software such as FreeCAD or Solidworks should also be an adequate alternative. The choice to specifically use Fusion360 aside from the license already being available is because of its parametric modelling feature.

This CAD design mentioned in [section 5.3.4](#) is illustrated in [Figure 5.7](#).



Figure 5.7: Block Box (full) CAD model

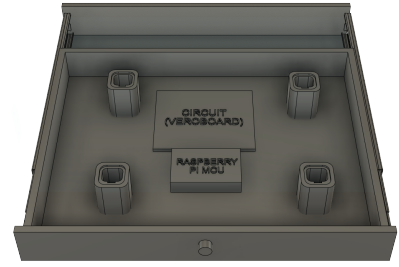


Figure 5.8: Block Box (interior) CAD model

Manufacturing Process

Block Box was handmade using waste chipboard wood pieces that was manually measured, cut, and assembled according to the CAD design specifications created in Fusion 360. This approach was chosen for its cost-effectiveness, environmental friendliness and the availability of materials. Therefore, this investigation has its element of promoting sustainable practices.

BLOCK BOX DIMENSIONS

The overall size of BLock Box was determined based on ensuring that the parcel in SP-01 could not fit comfortably but there would be enough space for other typical small package standards to fit as well.

Table 5.5: Dimensions of the Block Box and Inner Drawer Section

Component	Length (mm)	Width (mm)	Height (mm)
Outer Box	600	500	400
Inner Drawer	392	350	100

5.3.5 Bill of materials

Tab 5.6 gives a breakdown of the costs for each component.

Table 5.6: BLock Box BOM

Component	Description	Cost (ZAR)
Raspberry Pi 4B	Microcontroller	779.90
SD Card	16GB SD card for OS and software storage	75.00
Magnet Reed sensor	Door Sensor	24.00
Neodymium magnets	Door sensor trigger	30.00
HX711 Amplifier	Weight amplifier module	16.00
Load Cells (4x)	Weight Sensor	88.00
Solenoid Lock	Locking mechanism	139.00
4x4 matrix keypad	Access control mechanism	120.75
Power Supply Unit	Power supply	399.99
Piezo Buzzer	Sound feedback	36.90
Relay	Locking mechanism control enabler	26.00
Step-down converter	12V to 5V conversion for RPi powering	109.90
Cabinet Metal Runners	Rolling platform enabler	R149.00
Cabinet Metal handle	Aesthetic handle for user	11.00
Wood	Recycled waste material	0.00
Total		1930.44

5.4 Hardware Setup

5.4.1 System component overview

The Block Box system comprises of a range of crucial hardware components which can be seen in the table below:

Table 5.8: Hardware Components of the Block Box System

Component	Description
Wooden box	The structural housing of the prototype.
Raspberry Pi 4B	The central processing unit where software is executed on to manage hardware interactions. Additional components that are needed to setup MCU are mentioned in 5.4.2.
HX711 Load Cell Amplifier	An ADC that amplifies signals from load cells and feeds these to MCU.
Magnetic Reed Switch	Acts as a door sensor to detect whether the locker door is open or closed depending on the presence of a magnet on the door.
Solenoid Lock	Locking mechanism of the locker.
4x4 Matrix Keypad	Provides a user interface for the secure input of OTPs for access control purposes.

The interconnection of the components in Table 5.8 can be seen in Figure 5.9.

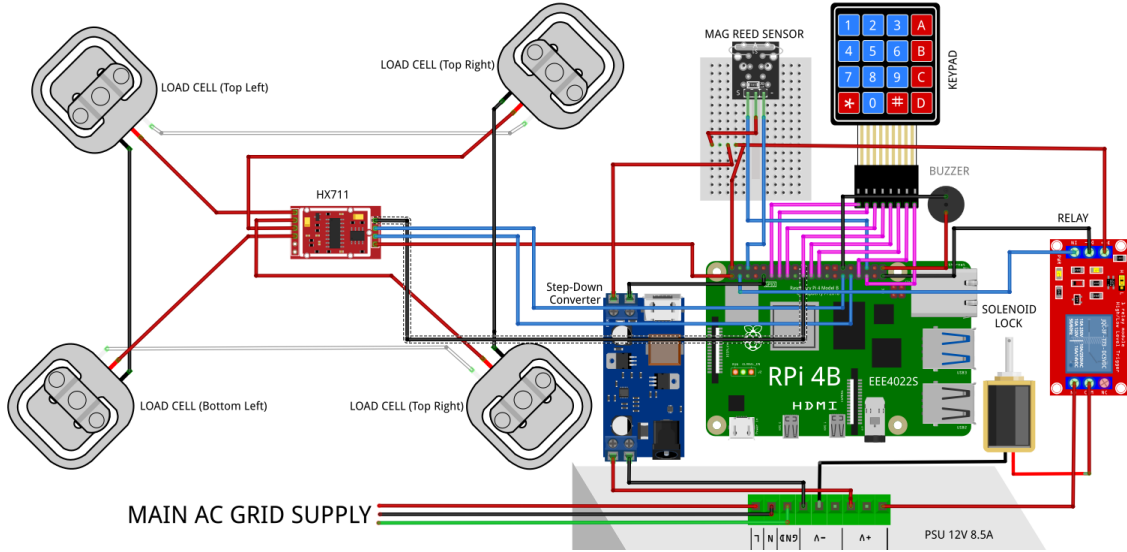


Figure 5.9: Interconnection of hardware modules in Block Box system

5.4.2 Microcontroller

As discussed in section 5.3.1 the RPi serves as BLock Box's microcontroller. The initial setup can be found in the [documentation](#). In summary, apart from the Raspberry Pi, one needs a microSD card (at least 16GB), mouse, keyboard, HDMI cable, micro HDMI adapter, a monitor to be able to setup a RPi and a USB-C power supply.

5.4.3 Weight sensor configuration

The components used for the weight sensor are the load cells as well as the HX711 amplifier module. The load cells are connected in a wheatstone bridge configuration as shown in Figure 5.9 and then wired as shown in Table 5.10.

Table 5.10: Wiring connections between the HX711 module and the four load cells

HX711 Pin	Load Cell Red Wire
E+	Top left load cell
E-	Bottom right load cell
A-	Top right load cell
A+	Bottom left load cell

Table 5.11: HX711 and RPi pin connection

HX711 Pin	Raspberry Pi 4B Pin
VCC	5V
GND	GND
DT	GPIO 5
SCK	GPIO 6

The HX711 was then connected to the RPi in as shown in [Table 5.11](#). However, it is important to mention that any of the GPIO pins can be used but the ones in [Table 5.11](#) are the ones that have been used in this investigation.

5.4.4 Solenoid lock integration

The solenoid lock is a straight forward component to wire up. It is the locking mechanism of the system and is connected to the RPi via a single channel relay module as seen in ??:

Table 5.13: Wiring connections between the relay module and Raspberry Pi

Relay Module Pin	Raspberry Pi GPIO Pin
VCC	5V
GND	GND
IN	GPIO 2

The solenoid lock is then connected to the relay module and the 12V psu [Table 5.14](#):

Table 5.14: Wiring connections between the relay module, solenoid lock, and power supply

Relay Terminal	Solenoid Lock Connection
COM (Common)	Positive lead of external power supply
NO (Normally Open)	Positive wire of solenoid lock
NC (Normally Closed)	<i>(Not connected)</i>
GND	Negative wires of solenoid lock and power supply

It is important to ensure that the operation of the lock is initially tested without being mounted on the door. This is because this solenoid lock is a fail-secure lock and there is always a risk of closing it accidentally and your system shutting down which means you would have to drill through your enclosure to get to the lock.

5.4.5 Keypad installation

The 16 different keys on a matrix keypad can be defined with the position of the respective row and column. This is why only 8 RPi GPIO pins are needed to fully define every input pin. The rows are the first four pins and the columns are defined as the next four each going left to right from R1,C1 to R4,C4 with the GPIO pin connections were illustrated in [Table 5.15](#).

Table 5.15: Keypad and RPi pin connection

Keypad Pin	Raspberry Pi 4B Pin
ROWS (R1, R2, R3, R4)	GPIO17, GPIO27, GPIO22, GPIO10
COLUMNS (C1, C2, C3, C4)	GPIO9, GPIO11, GPIO13, GPIO19

5.5 Software development

Once the hardware setup has been finalised it was time to move over to the software component. The software side of BLock Box was responsible for orchestrating the interactions between the different hardware components in such a way that a fully-functional system could be developed.

5.5.1 Programming Environment and project dependencies

As discussed in the design choices, python was the programming language of choice and five different classes were developed in order to be able to build BLock Box. The classes fulfilled different functions of the system and required the libraries and frameworks listed in [Table 5.17](#) to be installed, also preferably through a virtual environment on the RPi.

Table 5.17: Blockbox classes and dependencies breakdown

Class	Library	Description
FlaskServer	Flask	Web-server development class
BlockBoxGUI	Tkinter, PIL	Tkinter for GUI development and class uses PIL for image manipulation.
BlockchainIntegration	Web3	Blockchain interactions using python
OTPManager	PyOTP	OTP generation and verification
HardwareController	RPi.GPIO	GPIO pin control
TelegramHandler	Telegram	Telegram bot system integration

There are other software libraries that are not included in [Table 5.17](#) but they are used throughout the system and not necessarily class-dependent. An example of such a library which provides the system the ability to log and document its state while in operation. This is useful as Block Box is a logistics solution so being able to log the state of the system not just an added functionality but a necessity in terms of system security. Another library is the Threading module which allows the system to have multitasking capabilities. This for example allowed the program and the web server to be initialised concurrently. Furthermore, the os and the dotenv libraries were used for the secure management of the environment variables. These environment variables were personal information such as API_KEYS or the PRIVATE_KEY that the buyer inputted to sign off their transaction. This component of handling personal information securely fulfills SP-10.

Moreover, it is worth noting that although the full coded implementation is on the project [git](#), the description of all the methods of the classes in [Table 5.17](#) are listed in [section B.1](#). Most the the code is simple to follow and understand but he most complex and crucial piece of code has to be the part responsible for continuous monitoring the state of the system is [algorithm 1](#).

```

Function Monitor_System(stop_event):
  while stop_event is not set do
    // Update door status
    door_closed ← hardware.is_door_closed()
    if door_closed then
      | Update system_state['door_status'] to 'Closed'
    end
    Update system_state['door_status'] to 'Open'

    // Update item status based on weight
    weight ← hardware.read_weight()
    if weight > threshold then
      | Update system_state['item_status'] to 'Item placed'
      | Update system_state['item_in_box'] to True
    end
    Update system_state['item_status'] to 'No item placed'
    Update system_state['item_in_box'] to False

    // Pause before next check
    Sleep for 1 second
  end

```

Algorithm 1: System Monitoring Function

The various states of the BLock Box system and their initialised values are shown in [Table 5.19](#).

Table 5.19: System State Variables and Initial Values

door_status	item_status	item_col	price	tx_id	item_in_box	error_logs	tx_active
'Unknown'	'No item placed'	False	None	None	False	[]	False

Note: The update function of the states incorporate a threading lock(**state_lock = Lock()**). This Lock() implementation from the threading module ensures that only one thread has access to updating states at a particular time.

The listed pseudocode in [algorithm 1](#) clearly monitors the states of the door and the the status of the item in the box. All the other states do not need continuous monitoring and are updated through functional triggers.

Chapter 6

Methodology

The Block Box system comprises of a large hardware and software component to it. This methodology chapter, therefore, provides a detailed account of the processes, techniques and tools that were used in developing this Block Box system. Furthermore, there is a focus in detailing the methods used to test and validate the different areas which all contribute to the overall functionality of Block Box. The understanding in elaborating on a bit of the testing is to allow the system to be adequately reproduced and to establish a foundation for the results section which follows hereafter.

The adafruit-circuitpython-matrixkeypad library is the one used within this Block Box system as it simplifies the whole key pressing process. Moreover, once all connections have been made the test script keypadTest.py from the project [git](#) can be ran to unit test the component and ensure that the expected pressed key is showcased as a console output when pressed.

6.1 Validation and testing

There was a lot of testing that was conducted to ensure that each of the modules functioned as expected according to the project specifications listed in [Table 2.3](#). Moreover, once each element was tested in isolation the whole system was put together and a trail seller/buyer workflow was followed as a system integration test.

6.1.1 Unit testing

As partly mentioned in [section 5.4](#) test scripts were developed to individually test the hardware modules. It is worth noting that the unit testing of these hardware module and then later the software module is different to the ATP unit testing (UT) which is seen towards the end of this chapter. ATP unit testing is project specification based. The importance of making this differentiation is seen in the case where BLock Box has an OTP specification (SP-O2) but it does not have a Keypad specification even though the keypad is the interfaced used for the OTP. Therefore, for completeness all modules and specifications were tested.

Hardware

Solenoid Lock: The *lockTest.py* script was ran and developed for the lock unit test. This simple script toggles the GPIO pin of the relay connected lock consistently from HIGH to LOW in a timed manner to confirm the functionality of the lock.

Weight Sensor: The *loadCellCalibration.py*, *calibrationCurve.py* and *loadCellTest.py* scripts were ran to unit test the weight sensor.

The first script, *loadCellCalibration.py*, was used to attain the calibration factor using an iterative trial-and-error method. In this method an initial calibration factor of one was used and then the 1.034kg known weight was placed. The output of the sensor was recorded and the calibration factor was adjusted gradually until the error between the known weight and the measured weight is arbitrarily determined to be at an optimum minimum. Then the process is fine tuned using a different known weight.

The second script, *calibrationCurve.py*, the raw sensor output of five different known weights is plotted and the calibration factor is taken as the slope of the plot.

The last script, *loadCellTest.py*, takes in either one of these calibration factors and outputs the measured weight in a continuous loop. Thereafter, different known weights could be placed and graphs of the known weight vs actual weight could be plotted to measure the accuracy performance of the sensor.

Keypad: The *KeypadTest.py* script was used as a unit test which would simply output the pressed keys.

Door magnetic sensor: The magnetic reed sensor was unit tested using the developed *doorTest.py* script. This script configured the RPi GPIO pin of the magnetic sensor as an input pin and then continuously displayed its state in a looping fashion. Thereafter, a magnet would be brought towards the sensor and repelled and it was noted that a LOW indicated the presence of a magnet which could then be linked to the door being closed.

Software

Moreover, different software modules also went through a unit test to ensure that the units performed as expected before being combined with other coded blocks for the full system.

Telegram test: *botTest.py* was the script used to test the functionality of the telegram bot. The script simply initialised a bot using the TOKEN attained from telegram botFather. The chat ID is another requirement which was obtained via <https://api.telegram.org/botTOKEN/getUpdates> where a botTOKEN in the link was replaced with the given TOKEN from botFather. Note that a message would have to be sent to the bot from which ever device you have telegram installed as the API monitors communication to the bot which should show the chat ID. Upon initialisation a simple 'hi' message is sent from the bot to the chat ID as a test. The test was successful as the message arrived in the inbox of the device.

Flask web test server: The script *flaskTest.py* tested flask's functionality by printing 'Hi, flask is running' as an output on the flask web page that had been initialised with the default port and host='0.0.0.0'.

Blockchain transaction test: The blockchain part of the system was tested with a simple transaction using the developed *simpleTransfer.py* script. The script takes in two accounts and one of the accounts in which the money is to be sent from has to have a private key which is stored as an environment variable for security reasons. A specified ethereum amount is then transferred using Web3 modules from that account to the other one. The private key is needed to sign off the transaction as can be seen in the script.

6.1.2 Integration testing

The integration testing of block box system implies testing the system with everything connected together. This means that the following buyer/seller workflow should seamlessly be achieved:

System Transactional Success test

The full BLock Box system was tested in an integration test by simulating the ideal buyer-seller workflow ten times using a sample amount of R1,000.00 (0.22 Eth) as a means to test the system's overall success rate should the seller and the buyer interact with the system in an ideal manner.

Seller

The ideal workflow of the seller is shown in Figure 6.1 where the gray blocks indicate the physical interaction of the seller and the red flags are the required BLock Box system responses to the seller.

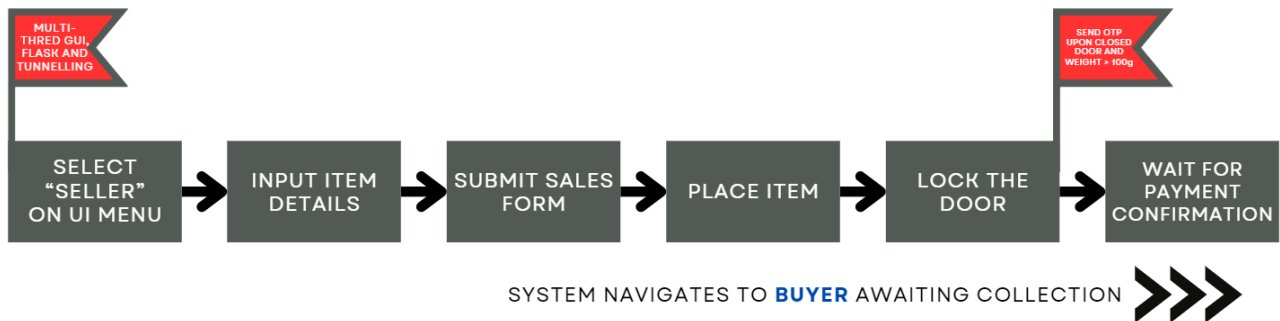


Figure 6.1: Seller Ideal Workflow

It is worth noting that the item details constituted of the following: Name, Description, advertised weight, Item price, the buyer's Ethereum address and an image of the item is uploaded to the system.

Buyer

The ideal workflow of the buyer is shown in Figure 6.2 where the gray blocks indicate the physical interaction of the buyer and the blue flags are the required BLock Box system responses to the seller.

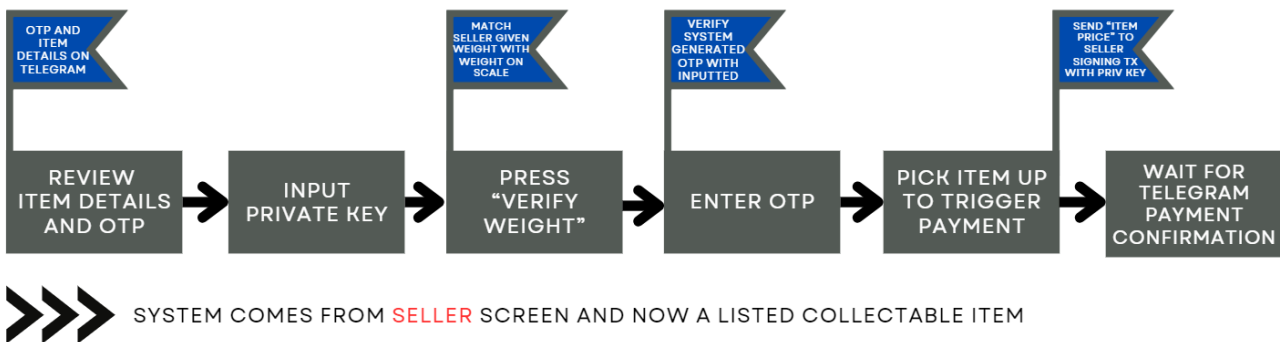


Figure 6.2: Buyer Ideal Workflow

As illustrated, successful OTP and weight verification unlocks the door, allowing the collect the item to take place. The system detects the item's removal and automatically processes the payment on the blockchain. This payment needs a private key to be about sign the transaction using `web3.eth.account.sign_transaction()` and then this signed transaction can be sent with method `web3.eth.send_raw_transaction()`. Exact implementation of how the Web3 library is used to send this transaction and how the transaction is prepared along with using Chainlink to do real-time price conversions is in the main code on the project [git](#).

System Verification

Throughout the testing process, the system continuously monitors the the state of the whole buyer/seller interaction. Therefore, the last part of this test would be to verify after all the trails that the contents of the system log file, the public web server and the JSON-formatted system state endpoint accessed with `ngrok_URL/system_system` all show a correct success message once a transaction is triggered.

Buyer interaction test

An additional integration test was conducted to simulate buyer interaction with the system during parcel collection. This focused on the weight sensor's responsiveness and its robustness under varying load conditions. A 1kg weight was placed on the scale to represent the parcel and the 120 seconds test was divided into specific intervals where different scenarios were applied as shown in [Table 6.1](#).

Table 6.1: Buyer Interaction Methodology test table

Time Interval	Description
0–30 seconds	The weight was left stationary on the scale to establish a baseline reading and ensure the system accurately detects a stable item presence.
30–60 seconds	Small, constant movements of the weight were undertaken to simulate disturbances a buyer might cause when inspecting the parcel without intending to remove it.
60–63 seconds	The weight was lifted off the scale for 3 seconds to simulate accidental pickups.
63–90 seconds	Aggressive movements were applied to the weight to represent vigorous inspection by the buyer.
90–120 seconds	The weight was left stationary again to observe if the system stabilises after disturbances and continues to monitor item weight correctly.

6.1.3 Acceptance Test Procedures (ATPs)

Acceptance Test Procedures (ATPs) are a set of test criteria that are developed to evaluate the project specification. It is, therefore, expected that each specification has at least one ATP. Furthermore, ATPs can be separated into UTs and UATs.

Table 6.5: Unit Test Procedures and Results

UT-ID	Unit Test Procedures (UTs)	SP Tested
UT-01	Description: Verification that a 450mm x 350mm x 160mm parcel fits into Block Box. Procedure: 1. Use a measuring tape to measure the interior dimensions of the Block Box. 2. Place a parcel of dimensions 340mm x 180mm x 80mm as per SP-01 into the Block Box. 3. Ensure the parcel fits comfortably. Acceptance Criteria: The interior section is 392mm x 350mm x 162mm, enough to fit the required box and leave a hand's length on each side so the box can be pulled in and out with ease.	SP-01
UT-02	Description: Ensure the OTP access control methodology works as expected. Procedure: 1. Run the main script <i>blockbox.py</i>. 2. Complete the seller form and press submit and wait to see if a random OTP is generated and sent to the buyer via Telegram. 3. Attempt to access the Block Box using both correct and incorrect OTPs. 4. Verify that OTPs expire after 10 minutes. Acceptance Criteria: OTPs are correctly generated, transmitted, expire after 10 minutes and access is granted only upon entering the correct OTP.	SP-02
UT-03	Description: Test the solenoid lock's minimum speed of engagement and disengagement due to the presence of a magnet. Procedure: 1. Connect the solenoid lock and magnetic reed sensor to their respective GPIO pins. 2. Run the script <i>accessControlResponse.py</i>. 3. Use a magnet to trigger the lock mechanism. 4. Measure the time taken for the lock to engage and disengage. Acceptance Criteria: The lock engages and disengages in less than 3 seconds.	SP-03
Continued on next page		

Table 6.5 – continued from previous page

UT-ID	Unit Test Procedures (UTs)	SP Tested
UT-04	Description: Confirm that the locking mechanism is a solenoid lock that locks when power is off. Procedure: 1. Visually inspect the locking mechanism and verify that it is a solenoid lock. 2. Power the lock and close the door. 3. Disconnect lock from power and ensure that the door locks and can only be unlocked should the solenoid be powered again. Acceptance Criteria: The locking mechanism is confirmed to be a solenoid lock through visual inspection and the user is unable to open the door when its closed and the lock is disconnected from the power supply.	SP-04
UT-05	Description: Ensure accurate detection of parcel presence via the weight sensor. Procedure: 1. Place known weights on the scale at various grid positions. 2. Run the <i>heatmap.py</i> script to record sensor readings and check that all grid positions detect a weight greater than 100g which is the threshold selected for the system to be able to detect the presence of a parcel. Acceptance Criteria: Sensor measurements have a margin of error less than 1%.	SP-05
UT-06	Description: Verify the embedded system’s microcontroller processing speed. Procedure: 1. Access the specifications of the microcontroller. 2. Confirm it meets or exceeds 1.2 GHz. Acceptance Criteria: The RPi 4B operates at a 1.5 GHz processing speed which is greater than the specified 1.2 GHz.	SP-06
UT-07	Description: Verify the embedded system’s RAM capacity. Procedure: 1. Access system information via software commands or system specifications and confirm the system has at least 1 GB of RAM. Acceptance Criteria: The RPi 4B is equipped with 1 GB RAM.	SP-07

Continued on next page

Table 6.5 – continued from previous page

UT-ID	Unit Test Procedures (UTs)	SP Tested
UT-08	Description: Validate IoT sensor data formatting and transmission to a global endpoint. Procedure: 1. Configure the IoT sensors to format data in JSON. 2. Transmit the data to a globally accessible endpoint. 3. Verify the integrity and correctness of the received JSON data on the endpoint. Acceptance Criteria: IoT sensor data is correctly formatted in JSON and successfully transmitted to the designated global endpoint without data corruption.	SP-08
UT-09	Description: Confirm the inclusion of the WiFi module. Procedure: 1. Run a connectivity test to verify the system can connect to the internet via WiFi. Acceptance Criteria: The WiFi module is inherently present in RPi 4B and the system maintains a stable internet connection.	SP-09
UT-10	Description: Test the weight sensor's accuracy up to 2 kg. Procedure: 1. Place a known weight under 2Kg and at 2kg on the scale. 2. Record the sensor readings. 3. Calculate the margin of error for each measurement. Acceptance Criteria: Weight measurements up to 2 kg have a margin of error less than 1%.	SP-10
UT-11	Description: Validate the verification system's accuracy. Procedure: 1. Perform multiple verification attempts to engage in a blockchain transaction to send funds from buyer to seller without any system errors. 2. Record the success rate on a per trail basis. 3. Calculate the overall accuracy percentage. Acceptance Criteria: The verification system achieves an accuracy greater than 95%.	SP-11

Table 6.6: User Acceptance Tests (UATs) and Results

UAT-ID	User Acceptance Tests (UATs)
UAT-01	Description: Ensure the system behaves correctly for a valid parcel delivery and has a smooth retrieval process. Procedure: Simulate the buyer/seller workflows listed in section 6.1.2 with a set parcel of a known weight and ensure that every step occurs without error. Acceptance Criteria: The door locks after the parcel is placed and the buyer is notified once the OTP is entered correctly. Buyer retrieves the parcel without issue and the seller receives their funds.
UAT-02	Description: Test the system's behavior when the seller inputs an incorrect parcel weight. Procedure: The seller enters a weight of 1kg for their parcel but places a different weight on the platform. Evaluate whether the system detects the discrepancy, prevents the door from locking to accept the parcel and ensures no collection notification is sent to the buyer. Acceptance Criteria: The system identifies the weight discrepancy, the door remains unlocked and no notification is sent to the buyer.
UAT-03	Description: Verify that the system handles parcels of varying sizes and weights correctly when placed at different positions on the scale platform. Procedure: Place boxes with parcel of different sizes and weights but not a parcel greater than 2kg on various parts of the platform. Observe how the system reacts in terms of weight detection accuracy and whether the door locks after the parcel is placed and the system proceeds as expected. Acceptance Criteria: The system does well to detect that something is on the platform in all cases but unless the parcel is placed in the center of the platform the whole system does not work as expected. Therefore, the door does not lock because of this inaccurate weight detection.
UAT-04	Description: Ensure that the system properly handles a case where the buyer might try and deceive the system by placing an item with the same weight. Procedure: The buyer retrieves the parcel and quickly swaps it for a fake item of the same weight. Acceptance Criteria: The system fails to detect the discrepancy that a different item has been inputted and the seller is notified to pick up their item even though this is the fake item the buyer sent.
UAT-05	Description: Test the system's distribution of quality user notifications delivery cycle. Procedure: Track the notifications sent via Telegram to both parties during different stages of the process and scrutinise that the level of detail that is provided aligns with the relevant stage. Acceptance Criteria: The system sends clear notifications at each relevant stage of the process and the buyers receive OTPs within a reasonable time window. Sellers are notified when the buyer retrieves the parcel.
Continued on next page	

Table 6.6 – continued from previous page

UAT-ID	User Acceptance Tests (UATs)
UAT-06	Description: Test the user satisfaction with the web interface of Block Box Procedure: Have the user access the system via a web interface to check the status of Block Box (e.g., locked, unlocked, or parcel weight). Ensure the user is able to understand everything on the server without confusion. Acceptance Criteria: The user can navigate the interface with ease and get relevant information such as the collection or door state statuses without additional technical assistance.
UAT-07	Description: Have the user try to forcefully open the prototype without any thermal or power tools to try and simulate the stress impact a thief would place on the locker box to try bypass the system illegally and check the safety of the wooden prototype. Procedure: The user has to ensure that the door is locked and then pull as hard as they can on the 3D printed handle of the prototype to try and open the box. The average user should not be able to open the box using their bare hands without power tools. Acceptance Criteria: The wooden box is sturdy and does not open easily. The handle managed to break off when being pulled on but the box remained fully intact.

Chapter 7

Results

This chapter aims to expand on the methodology section, [chapter 6](#), by illustrating both of the qualitative and quantitative results that were obtained from the mentioned testing and validation procedures. These tests include some of the mentioned UTs as well as documented results of the UATs that were computed on a per trail basis. The chapter is organised in an order that aligns with the way the testing procedures were illustrated.

7.1 Prototype Development

The handmade wooden prototype that was made to fulfil SP-01 is illustrated in [Figure 7.1](#). This prototype serves as the physical enclosure and housing of the BLock Box system.



Figure 7.1: Block Box wooden prototype

It was noted during user testing that the box's interior open area of 467 mm $\times$ 350 mm $\times$ 162 mm provided ample space for parcels of dimensions specified in SP-01 to easily be placed within the system. Moreover, similar ranged parcels of 250 mm $\times$ 220 mm $\times$ 60 mm and 300 mm $\times$ 200 mm $\times$ 100 mm fit and worked just as well with the BLock Box system. The packaging used for testing was placed on a kitchen scale and all had an expected weight of just over the weight detection threshold of 100g used within the BLock Box system.

7.2 Unit Testing Results

7.2.1 Hardware Components

Solenoid Lock Testing

The *accessControlResponse.py* script verified the functionality of the solenoid lock and the lock engaged and disengaged reliably within 1 second. This was well under the specified maximum of 3 seconds listed in SP-03. The fastest switching speed that could be recorded with a handheld stopwatch was 0.14s. Moreover, the qualitative observations from the testing indicated smooth operation without the lock getting stuck. However, after a few minutes of constant switching the lock had a detectable increase in heat.

Door Magnetic Sensor Testing

Using the *doorTest.py* script, the magnetic reed sensor accurately detected the door's open and closed states. The detection distance between the sensor and the magnet was found to be approximately 2cm.

Keypad Testing

The *KeypadTest.py* script confirmed that all keys on the keypad were functional with no instances of misinterpreted inputs during the test. The tactile feedback provided by the keypad was a satisfactory qualitative observation that added help in knowing exactly when a key was pressed.

Weight sensor performance evaluation

Kitchen Scale Results



Figure 7.2: Known Weights used for calibration



Figure 7.3: Known Weights with used kitchen scale

The known weights used in the calibration process that were initially placed on a kitchen scale as seen in [Figure 7.2](#) and [Figure 7.3](#) deviated from their labeled manufactured mass values as illustrated in [Table 7.1](#).

Table 7.1: Kitchen scale known weight results

Labelled mass (kg) on weight	Kitchen Scale result (kg)	% Error
0.5	0.504	0.8
1.0	1.034	3.4
1.0	1.052	5.2
2.5	2.619	4.76
2.5	2.492	0.32
2.5	2.516	0.64
2.5	2.466	1.76

Upon knowing the accurate values of the utilised known weights the weight sensor was then calibrated using two methods to ensure precise measurement as per SP-05. These methods are the iterative trial-and-error and calibration curve analysis methodologies as described in [section 6.1.1](#).

Calibration factor determination

Iterative trail and error: This method took 30 trails to be able to come to an optimum calibration factor of -21887.67 using the 1.034kg and 0.504kg weights as the two calibration masses. The factors that were considered in this iterative test can be seen in [Figure 7.4](#).

Table 7.2: Factors considered in the Iterative trail and error calibration test

Trails	Calibration Factor	Known Weight	Measured Weight	% Error (abs)
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[Figure 7.4](#) showcases this experimentation process where the solid red is the error between the known weight and measured weight. The dotted red line is the 1% error line which is the SP-05 requirement and the goal is to keep changing the calibration factor (yellow) until the solid red line is below the dotted red line. The blue line in [Figure 7.4](#) just shows the known weight that was used at on that specific trail between the 1.034kg and 0.504kg.

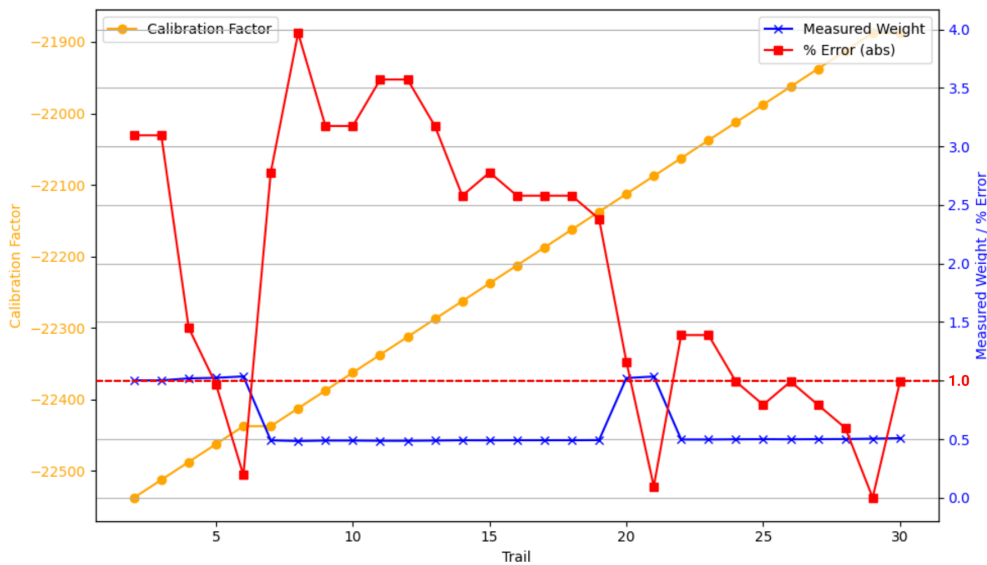


Figure 7.4: Iterative method calibration data

Calibration curve analysis:

The slope of the second calibration methodology as seen in Figure 7.5 came out to be -25072 which was the used calibration factor. The maximum known weight that was used in this method was 2.492kg compared to the 1.034kg in the first method. The -25072 was also the calibration factor used from this point onwards in the investigation.

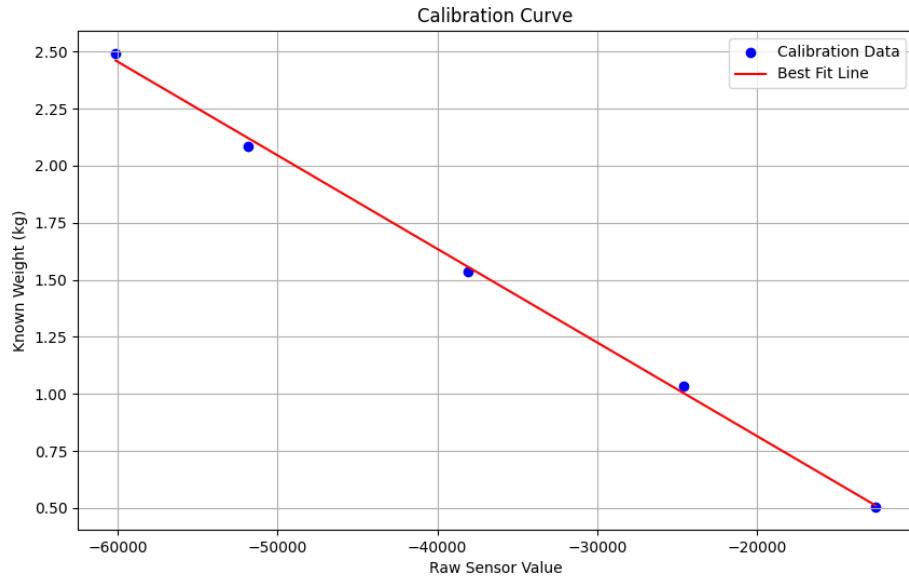


Figure 7.5: Calibration curve calibration data

The quantification of the accuracy of the weight sensor

In using the -25072 calibration factor, the actual weight against the measured weight of the five known weights are plotted in Figure 7.6. Moreover, the same test where three samples per known weight each a second apart are computed and the measurement errors are plotted as seen in Figure 7.7.

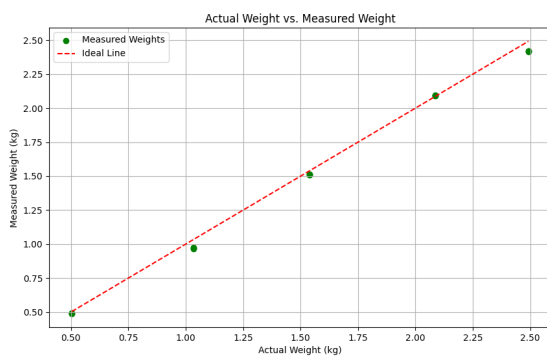


Figure 7.6: Actual weight vs measured weight for -25072 calibration factor

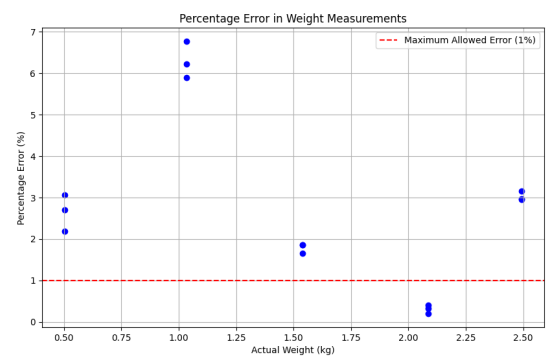


Figure 7.7: Multiple samples measurement error for multiple known weights

Qualitative Observations

Table 7.3 illustrates the qualitative observations during the unit testing of hardware components.

Table 7.3: Qualitative Observations from Unit Testing

Observation	Result
User Interface	Clear layout with easy to follow instruction.
System Responsiveness	Quick system feedback on user inputs.
Reliability	No hardware or software failures occurred during testing.

7.2.2 Software Components

Telegram Bot Functionality

The *botTest.py* script managed to get the configured bot to send a ‘hi’ message to the chatID of a smartphone-installed telegram app.

Flask Web Server Testing

The *flaskTest.py* script successfully launched a local web server showcasing the prompt ‘Hi, flask is running’.

7.2.3 OTP Authentication Accuracy

The OTP authentication system was tested over 30 trials to ensure secure access control.

Table 7.4: OTP authentication test results

Correct OTP Entry:	10/10
Incorrect OTP Entry:	0/10
Expired OTPs:	0/10

As seen from Table 7.4 the quantitative data backs the qualitative observation of the OTP system being simple and reliable.

7.3 Integrated Testing results

7.3.1 System Transactional Success Test

A bag of 1kg rice was arbitrarily said to costs R1 000.00 (0.022 ETH at the time this report is compiled). The item was the one used to test 25 trails of the system transaction workflows seen in Figure 6.1 and Figure 6.2. The metric of this integrated test can be seen in Table 7.5.

Table 7.5: Parcel Verification Test Results

Door Test	OTP Test	Weight Verification	Server Test	Transaction Success
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96% of the 25 tests listed in Table 7.5 were successful as the only test that failed was because of a single fail in the weight verification checkpoint of that trail.

Qualitative Observations from Transactional test

Qualitative observations during the integrated transactional tests can be seen in [Table 7.6](#).

Table 7.6: Qualitative Observations from Integration Testing

Observation	Result
User Experience	Clear layout with easy to follow instruction.
Notification System	Telegram interface provided well-structured, real-time, concise notifications throughout the whole testing.
System Stability	No unexpected errors or system instability was observed.

7.3.2 Buyer Interaction Test

Unfiltered Interaction

The buyer interaction test with the specific testing scenario illustrated in [section 6.1.2](#) produces the blue unfiltered weight output in [Figure 7.8](#) when the 1.034kg known weight was the item under test. The red dotted line is the 0.1kg threshold where the transaction triggers if the weight goes below it. The vertical green dotted lines segment the different phases of the testing scenario where different loading conditions apply.

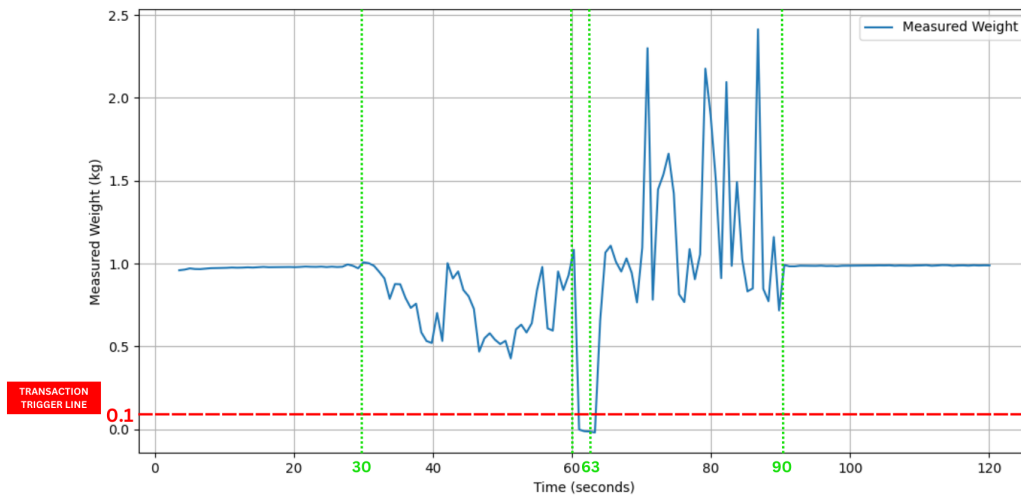


Figure 7.8: Unfiltered buyer interaction test

Median filter application to delay trigger of payments

Median Filter Application

A median filter was applied to the sensor data to stabilise the aggressive fluctuations of the readings to reduce possible false transaction triggering during buyer interaction.

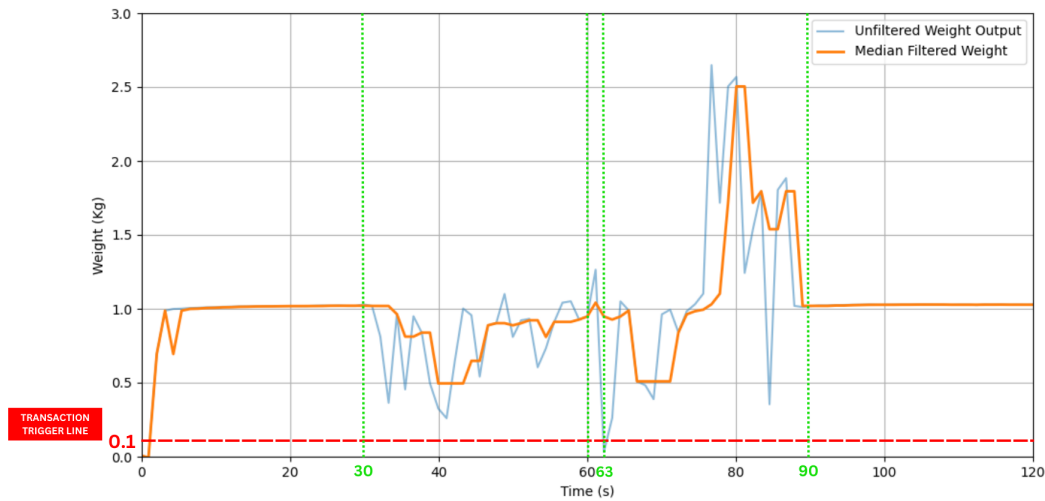


Figure 7.9: Median filter integration to buyer interaction test

Qualitative Observations from Buyer Interaction Test

Observations from the buyer interaction test are shown in [Table 7.7](#).

Table 7.7: Qualitative Observations from buyer Interaction Test

Observation	Result
Parcel Handling	The system took longer to send a payment receipt with the median filter application when the item was lifted off the platform.
System Feedback	The system was quick to update its state when a parcel was removed or placed.

7.3.3 Off-Centre Placement of Parcel

The effect of the positional placement of a parcel by the user on the scale's accuracy was evaluated and shown in [Figure 7.10](#). The illustration shows a 3x3 heatmap of the measurement errors that were conducted at those specific areas on the BLock Box platform.

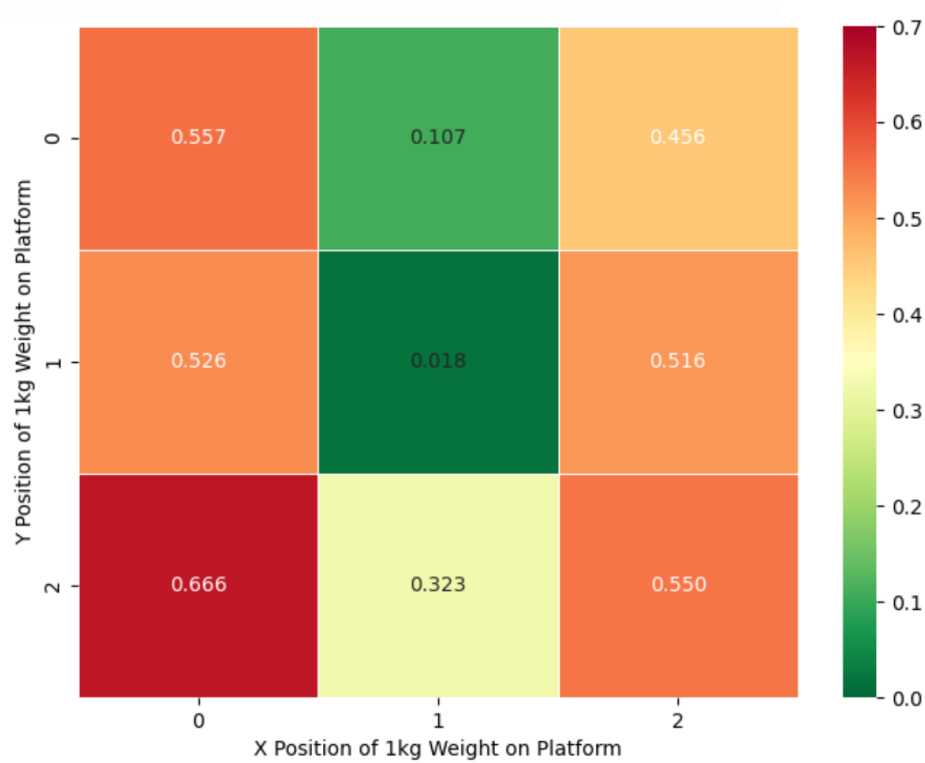


Figure 7.10: Interaction Heatmap

7.4 Acceptance Test Procedures (ATPs) Results

The summarised results of the Unit Tests (UTs) and User Acceptance Tests (UATs) can be seen in [Table 7.8](#).

UT-ID	Pass/Fail	UAT-ID	Pass/Fail
UT-01	Pass	UAT-01	Pass
UT-02	Pass	UAT-02	Pass
UT-03	Pass	UAT-03	Fail
UT-04	Pass	UAT-04	Fail
UT-05	Fail	UAT-05	Pass
UT-06	Pass	UAT-06	Pass
UT-07	Pass	UAT-07	Pass
UT-08	Pass	-	-
UT-09	Pass	-	-
UT-10	Fail	-	-
UT-11	Pass	-	-

Table 7.8: Unit and User Acceptance Tests Results

Chapter 8

Discussion

This chapter provides an in-depth analysis of the results presented in [chapter 7](#). The findings are interpreted in the context of the BLock Box system's objectives and the specifications outlined in [Table 2.3](#). Moreover, this interpretation is then also related to concepts found in [chapter 4](#), the literature, to make the overall discussion a coherent and concise one.

8.1 Unit testing

8.1.1 Weight Sensor Performance Evaluation

The accuracy of the weight sensor is a crucial metric for the overall success of the verification system. This metric directly impacts SP-05 and SP-10 which require the weight sensors to have a margin of error that is less than 1% when measuring the mass of objects up to 2 kg.

Comparison of calibration methods

The iterative trial-and-error method produced a calibration factor of -21887.67 and the calibration curve analysis method one of -25072. This vast difference in the two calibration factors was mainly because of that the iterative method only made use of two known weights whilst the slope approach made use of five different known weights. Moreover, the iterative method takes the approach of only finding the calibration factor on one weight at a point in time. Therefore, since the calibration curve method used multiple known weights it should provide a more sound and statistically robust calibration as it should not be bias to a specific weight. This was the justification to use -25072 as the calibration factor throughout the system.

Accuracy Evaluation

The results showcased in [Figure 7.7](#) show that the weight sensor that was calibrated using the slope calibration method did not achieve a margin of error that is less than 1% for most of the measured cases except for the 2.086 kg known weight case. 4/5 of the tests were at least 3% or less which could still constitute as a negligible error. Moreover, it also suggests that achieving an error rate that is consistently 1% or less is possible but different calibration strategies would have to be implemented.

Moreover, a robust platform positioning strategy has to be applied as the system's accuracy deteriorates poorly should the parcel not be placed in the dead-centre of BLock Box. A strategy that was trailed was applying correction factors to try and have a *proportional control* approach in fixing the observed error which can be seen in [Interaction Heatmap AFTER correction](#).

Initially the thought was that this approach works as the presented heatmap is a lot more green with a lower error per unit grid than the original [Figure 7.10](#) which illustrates an improvement of the overall system error rate. However, this approach does not work as the sensors are not precise enough. There is just too much error that seems to be non-linear that is embedded within the calculated correction factors that this approach did not work when switched over to another weight. Therefore, another approach all together has to be looked into to try and fulfill SP-05 and SP-10.

8.1.2 OTP Authentication Reliability

The OTP authentication system demonstrated that it was 100% reliable during its unit testing results shown in [??](#). This result confirms that the OTP mechanism effectively secures access to the BLock Box, the fulfillment of SP-02 as the OTP system secures access to BLock Box securely.

8.2 Integration Testing Analysis

8.2.1 System Transactional Success

The integration testing of the buyer-seller workflow yielded an 96% success rate as shown in [??](#). The in the weight verification are most likely because of a slightly off-center parcel placement as this correlates with the SP-05 failure. This sensitivity could lead to user frustration and reduced trust in the system if transactions fail simply because a user placed their parcel towards an edge of the box and not in the center.

8.2.2 Off-Center Parcel Placement Challenges

These tests revealed a lot more about the system than just a basis to say SP-05 and SP-10 have failed because the errors are not within the required 1% range. One such insight is the indication that the positioning of the platform and the weights may not be fully symmetrical and flat. This is because if that was the cause the errors on the left side would be equal on average to the errors on the right hand side but they are not as they are greater.

8.2.3 Buyer Interaction test

The buyer interaction test that can be seen in [Table 7.7](#) provided a lot of insight over how the weight sensing of the BLock Box system would respond when a buyer is busy manually verifying their item. This insight is best evaluated on a per interval basis as shown in [Table 8.1](#)

Time Interval	Insight
0–30 seconds	The weight was left stationary on the scale to establish the steady state reading and ensure the system accurately detects a stable item presence and accurate weight.
30–60 seconds	This phase tested the system’s ability to handle slight fluctuations without falling below the threshold blockchain payment triggering line.
60–63 seconds	This phase is arguably the most important one as it checks whether the system would prematurely process the transaction when the item is briefly picked up. This case is most likely to occur in real-life as the buyer would want to turn the item over and in doing this lift it up. It can clearly be seen that the red line in Figure 7.8 is crossed in this stage which indicates a flaw that the blockchain transaction would be triggered even if the pick up from the buyer was accidental. This problem, however, was fixed with a median filter in Figure 7.9 but at a security trade-off. This is because the instant triggering was designed so that buyer’s cannot succeed when trying to beat the system by swapping the item with a worthless item with the same weight. This is now a possibility with the filter but at the benefit of an enhanced user verification experience.
63–90 seconds	The aggressive moment in this phase tested the system’s capacity to handle high frequency disturbances.
90–120 seconds	The weight was left stationary again to see if the system settles at its intended steady state point.

Table 8.1: System Testing Methodology

8.3 Acceptance Test Procedures Review

The two UTs that failed were already discussed and elaborated, however, not so much on the failed UATs. Therefore, [Table 8.2](#) elaborates on these failures.

Table 8.2: Summary of UAT Failures

UAT-ID	Issue Description
UAT-03	The system was unable to accurately detect the weight of parcels placed off-center as shown by the high errors in Figure 7.10 . This UAT failed not because of the different boxes but the off center placements. The size of the box was seen to be not too much of a big factor as long as it weighs around the same as a standard small box.
UAT-04	The implementation of the median filter was the cause of this UAT failing as the user is able to swap out the weight for an identical one should it be done within 3 seconds.

Furthermore, the results of the ATPs indicated an overall success rate of 81.8% for the UTs and 71.2% for the UATs as illustrated in [7.5](#). The failures of the UTs and UATs can be attributed to solvable issues such as finding a more suitable calibration methodology to improve the accuracy of the weight sensor.

Chapter 9

Conclusions

9.1 Summary of Findings

The purpose of this project was to design, develop, and evaluate the BLock Box system, a secure, user-friendly smart parcel locker solution, that aims to solve the asymmetric information last-mile delivery payment problem. This problem was solved to through the integration of blockchain technology with the weight sensing IoT technology.

This report began with an introduction to this problem back in ???. Here the motivation for the project alongside the scope and objectives of the project was established. The [chapter 2](#) touched on the specific requirements to be achieved throughout the investigation.

Moreover, since blockchain technology is still a very niche area of research [chapter 3](#) was written next before introducing the literature in [chapter 4](#) where a comprehensive examination of existing solutions and technologies that are relevant to the project were outlined.

The bulk of the work for this project was detailed in [chapter 5](#), which described the undertaken design and implementation process. This included all the design choices of BLock Box's four established sub-modules as well as the system architectural design.

[chapter 6](#) then provided a clear roadmap for replicating and understanding the system's development process. In this section all the testing and validation procedures were established.

In [chapter 7](#), the quantitative and qualitative results of the testing procedures mentioned were then presented. These results were then finally discussed and interpreted in [chapter 8](#).

In summary, the project achieved its main objective of illustrating that a blockchain-based transaction can be triggered using IoT technology. In doing so this is a concept project that solves the asymmetric information problem while also leaving a lot of room to scale the project to ensure its robustness.

9.2 Concluding Remarks

The development of the BLock Box system represents a significant contribution towards the enhancement of parcel lockers within the last-mile delivery framework. By leveraging blockchain technology, the project not only makes peer-to-peer transactions convenient and safe but solves a real-world economical problem.

Chapter 10

Recommendations

Although the Block Box system is already a huge step in the right direction for the last-mile delivery process, there are a few areas where further improvement can be imposed. These developments would potentially make the system even more user-friendly and secure.

10.1 Modular design approach

The biggest advantage of designing a system in a modular manner is that the underlying system then becomes extremely scalable. Therefore, as the demand for last-mile delivery solutions continue to grow it is imperative that the Block Box system is scalable in order to accommodate large parcel volumes. Although, modularity is not the only way to achieve scalability, designing a modular design for the lockers would allow them to be easily expanded or reconfigured based on the underlying demand.

10.2 Enhancing User Interface and Experience

Improving the user interface (UI) and user experience (UX) is crucial for user satisfaction and adoption of the BLock Box system. This could imply that the web interface be enhanced with real-time tracking features or as simple as providing more detailed notifications for the involved parties. Furthermore, ensuring that the system is accessible to all, including those with disabilities, should be of utmost importance.

10.3 Improved Chainlink integration

It is worth noting that the v1 version of Chainlink used JSON formatting but the system recently moved to TOML and provided tools in their [documentation](#) for this migration. TOML is a superset of JSON and has even simpler and stricter syntax which makes errors even less. Therefore, BLock Box should look into formatting the API endpoint in TOML for future Chainlink integrations.

Moreover, the LinkWells [6] is an oracle provider startup which could be useful for future developments of BLock Box. This is because they provide a free pre-deployed Chainlink smart contracts which can be used to pull data from any public HTTPs site and put it on chain.

10.4 Enhancing Weight Measurement Accuracy

To address the concern of not getting below 1% for the weight measurement accuracy error better calibration methods can be employed to try and improve the accuracy. Firstly a lot more known weights with more precise load cells should be used with existing strategies. Then upon reevaluation, machine learning algorithms can be employed to learn from a database of common objects and their weights and adjust the calibration to be bias to weights in the expected region before measuring.

10.5 Structural improvement

Although this version of BLock Box is made with wood a structural improvement is changing the material and perhaps applying for a baseline safety compliance certificate to enhance user satisfaction and trust. A suggestion is making the locker based off of UL Solutions' UL RSC Level I certification. This locker system must, therefore, be steel based with walls greater than 18.5cm thickness and a door thickness greater than 0.5cm. This UL RSC level 1 standard is adequate for the improved version of BLock Box as it is built for burglars who are likely to try infiltrating the system with basic tools for about 5 minutes. UL Solutions' other safety standards are listed in [62].

10.6 Conclusion

The outlined recommendations address key areas for improvement within the BLock Box system. Implementing these enhancements will improve scalability, security, accuracy, and user satisfaction, essential for the system's success in real-world applications and for meeting the evolving demands of last-mile delivery solutions.

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Appendix A

GA Analysis Table

GA	Requirement	Justifications with section in the report
1	Problem-solving	In this investigation, problem-solving is demonstrated in chapter 3 where a niche topic like Blockchain technology was broken down and explained. Furthermore, this IoT and Blockchain integration is still a novel topic so the frameworks and the implementation provided in chapter 5 are the epitome of critical problem solving. Even finding literature on this topic was a problem that was solved.
4	Investigations, experiments and data analysis	The investigation is reflected in chapter 7 . The plots that were produced show a high level understanding of being able to give quantitative metrics for a methodological procedure. Furthermore, chapter 4 showcases a high quality investigation into the principals governing BLock Box.
5	Use of engineering tools	The use of modern and archaic engineering tools can be found in chapter 5 . The old tools being the woodworking that was computed and the new tools being using tools such as laser cutting, 3d printing, fusion 360 and then the software element of intensive python back and front end programming.
6	Professional and technical communication (Long report)	This very report serves as evidence for the completion of this GA. Furthermore, weekly meetings with supervisor was conducted.
8	Individual work	Most of the frameworks that are shown like the sequence diagram in chapter 5 and , the thinking behind triggering payments with item pick ups, flow diagrams throughout the report are all illustrations of my own individual work.
9	Independent learning ability	When venturing into the realm of a topic that is niche, individual work comes as a necessity. The understanding of Blockchain technology showcased in chapter 3 is an example of the hours taken to self-learn a topic outside of my degree.

Appendix B

Code

B.1 Software per class method description

```
1 class TelegramHandler:
2     def __init__(self, token, chat_id, role): #Initialisation of telegram bot with details that were
3         ↪ attained on telegram via botFather.
4     def send_message(self, message): #Sending specified message to specified chat.
```

Listing B.1: TelegramHandler Class Code Workings

```
1 class HardwareController:
2     def __init__(self): #GPIO pin config (BCM numbering)
3     def setup_gpio(self): #Lock and door GPIO setup
4     def lock_door(self): #Solenoid pin set to LOW to lock door
5     def unlock_door(self): #Solenoid pin set to HIGH to unlock door
6     def is_door_closed(self): #Cheaking if magnetic reed sensor is LOW
7     def read_weight(self): #outputing get_weight result of HX711
8     def cleanup(self): #GPIO settings cleanup on exit
```

Listing B.2: HardwareController Class Code Workings

```
1 class OTPManager:
2     def __init__(self, secret_key, valid_duration=OTP_TIMEOUT): OTP functionality initialisation
3     def generate_otp(self): # Unique OTP creation using totp.now()
4     def verify_otp(self, otp_entered): # OTP verification using result from totp.verify
5     def is_otp_expired(self): # Checking creation time of OTP and current time and making sure that
6         ↪ expired means this time is greater than set valid_duration
```

Listing B.3: OTPManager Class Code Workings

```

1 class FlaskServer(Thread):
2     def __init__(self, app, host='0.0.0.0', port=5000): # Flask web server set up on default port and
        ↳ to listen to external devices
3     def run(self): Method to start server

```

Listing B.4: FlaskServer Class Code Snippet

```

1 class BlockchainIntegration:
2     def __init__(self): Blockchain settings initialisation and ethereum network connection via INFURA
        ↳ node provider
3     def get_eth_price_usd(self): # Using Chainlink pricefeed smart contract to get ETH from USD
4     def get_zar_usd_rate(self): # Using exchange rate API to get ZAR from USD
5     def calculate_eth_amount(self, zar_amount): # Uses previous two methods to go from ZAR -> ETH
6     def set_transaction(self, buyer_address, item_price_zar): # Verifies if ETH balance of buyer
        ↳ exceeds item price.
7     def trigger_payment(self, buyer_private_key, item_price_zar): #Using Web3 methods to sign and send
        ↳ the transaction valued at the item price

```

Listing B.5: BlockchainIntegration Class Code Workings

```

1 class BlockBoxGUI:
2     def __init__(self, root, hardware, buyer_bot, seller_bot, otp_manager): #GUI initialisation
3     def setup_gui(self): #Layout of opening interface configured
4     def create_intro(self): #Introductory seller/buyer screen displayed
5     def open_seller(self): #Navigation to seller form where item details are inputted
6     def create_seller_form(self): #Seller form page layout
7     def upload_image(self): #Allowing seller to upload image of item from memory using .open(path)
8     def submit_seller_data(self): #Checks if submit button is pressed and sets up transaction based on
    ↪ inputted data
9     def send_otp_via_telegram(self, otp): #Created otp upon submission sent as message on telegram
10    def generate_transaction_id(self): #Random id created per transaction
11    def save_seller_data(self):
12    #Inputted data with ID from seller saved to be shown to buyer
13    def wait_for_door_close(self):
14    #Loop that only terminates when door is closed
15    def wait_for_door_open(self):
16    #Loop that only terminates when door is open
17    def wait_for_item_placement(self):
18    #Checking when weight sensor detects an item greater than 100g
19    def open_buyer(self):
20    #Navigation to buyer page
21    def display_seller_data(self):
22    #Item details to be shown to buyer
23    def verify_weight(self): #Seller inputted weight matched to weight on the platform
24    def read_keypad_input(self):
25    # Returning pressed keys for OTP
26    def monitor_item_collection(self):
27    # Looping function to consistently check status of item on scale
28    def send_seller_message(self, msg): #Sending seller messages to telegram chat
29    def notify_buyer_otp_expired(self): #Alerting the buyer via telegram if OTP is expired
30    def reclaim_item(self): #Alerting seller to collected rejected item
31    def reset_system(self): # Taking system back to initial state
32    def clear_buyer_gui(self): # Elements in buyer GUI resetted
33    def clear_seller_gui(self): # Elements in buyer GUI resetted

```

Listing B.6: BlockBoxGUI Class Code Workings

Appendix C

Additional Files and Schematics

C.1 Application of correction factor for weight accuracy

After correction factors

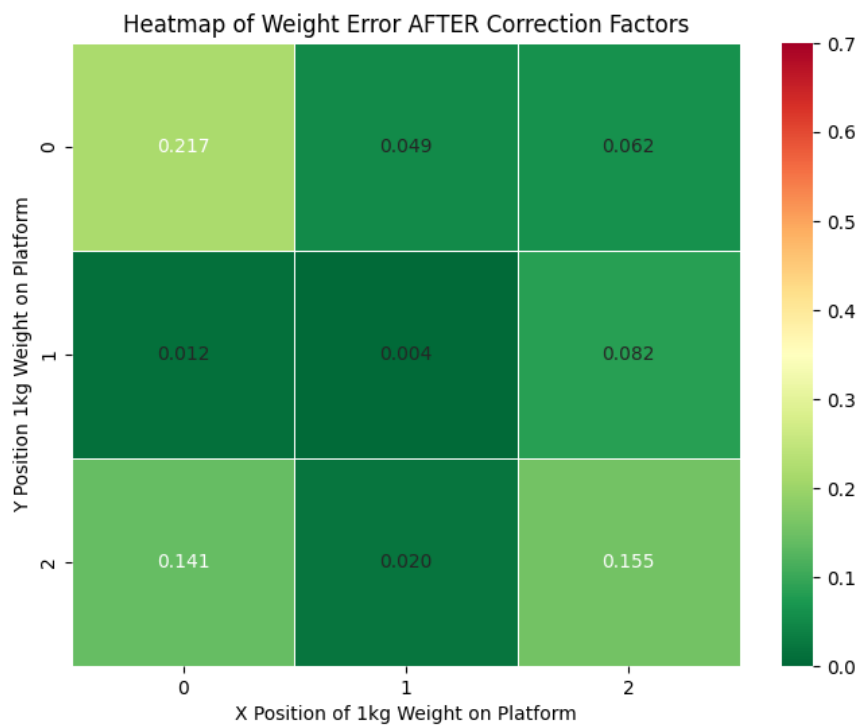


Figure C.1: Interaction Heatmap AFTER correction

C.2 GUI Interface

Intro page:

Seller Page:

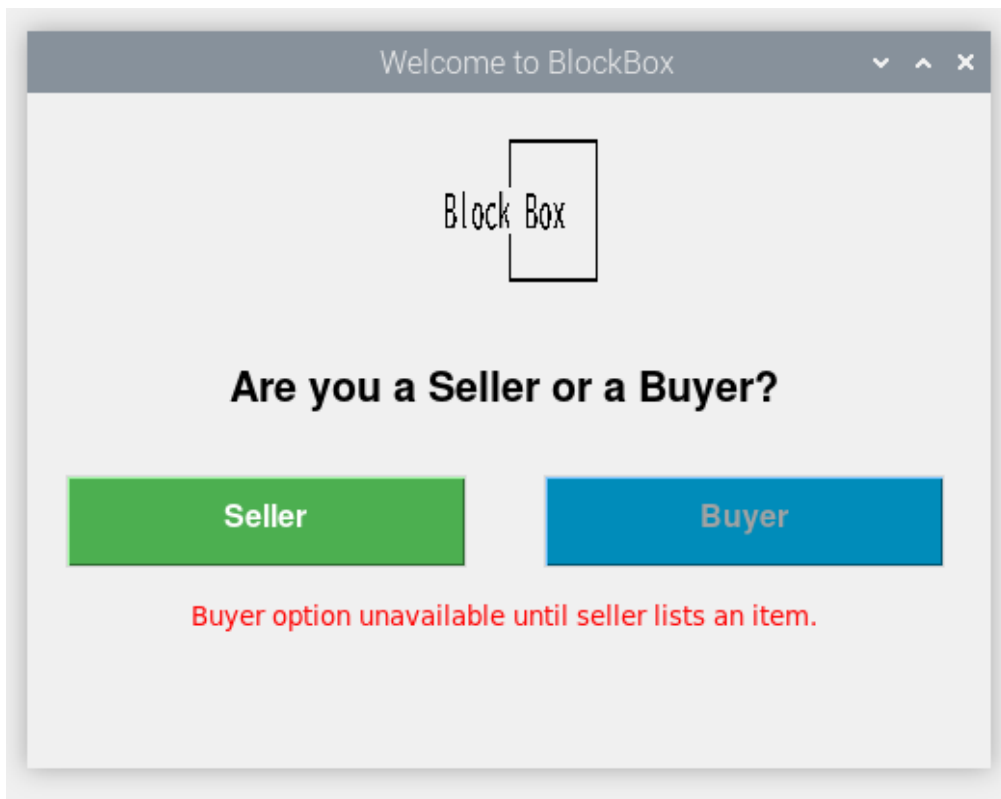


Figure C.2: Intro page

A screenshot of a web application window titled "BlockBox System". The window contains a form for listing an item. The form has five input fields: "Item Name:", "Description:", "Advertised Weight (kg):", "Price (Rands):", and "Buyer Ethereum Address:". Below the "Buyer Ethereum Address:" field is a blue button labeled "Upload Image". At the bottom of the form is a green button labeled "Submit".

Figure C.3: Unpopulated Seller page

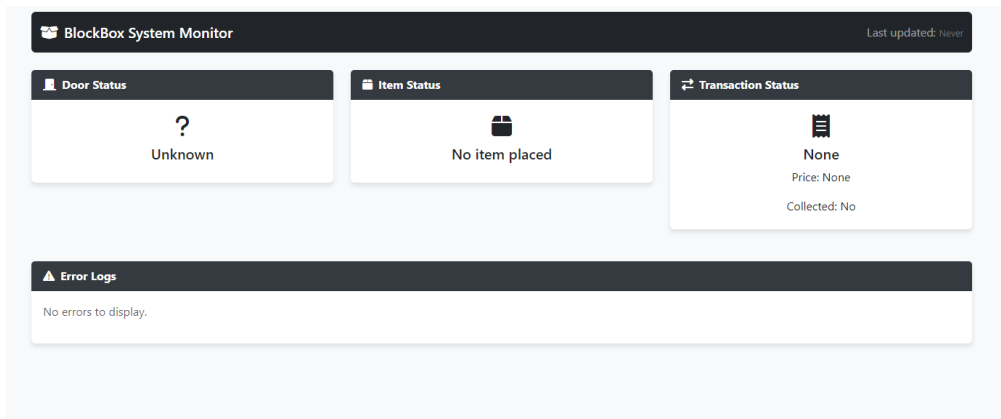


Figure C.4: Uninitialised Web Server

C.3 Web Server

Seller Page(Populated):

 The screenshot shows a web form titled "BlockBox System" with the following fields: "Item Name:" with the value "Iphone 15"; "Description:" with the value "1TB Storage | Bargain | Best smartphone in the world"; "Advertised Weight (kg):" with the value "0.176"; "Price (Rands):" with the value "20000"; and "Buyer Ethereum Address:" with the value "0xD6453762B3F13996f0F1a890a05abeEB261c782A". Below the form is an "Upload Image" button and a placeholder image of an iPhone 15. At the bottom is a green "Submit" button.

Figure C.5: Populated Seller page

Note that feel of buyer page is similar to this.

C.4 Prototype images

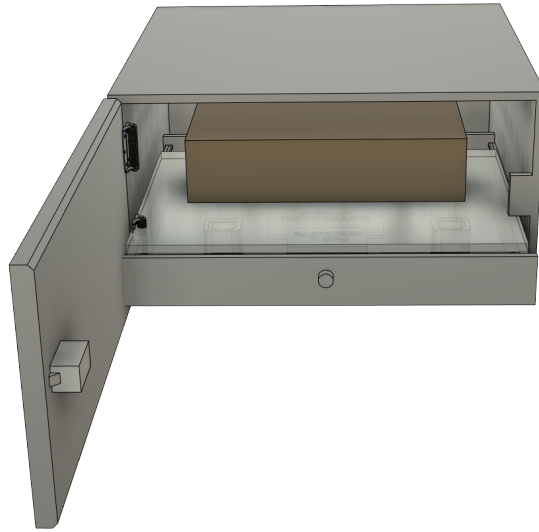


Figure C.6: Box fitting in simulated version

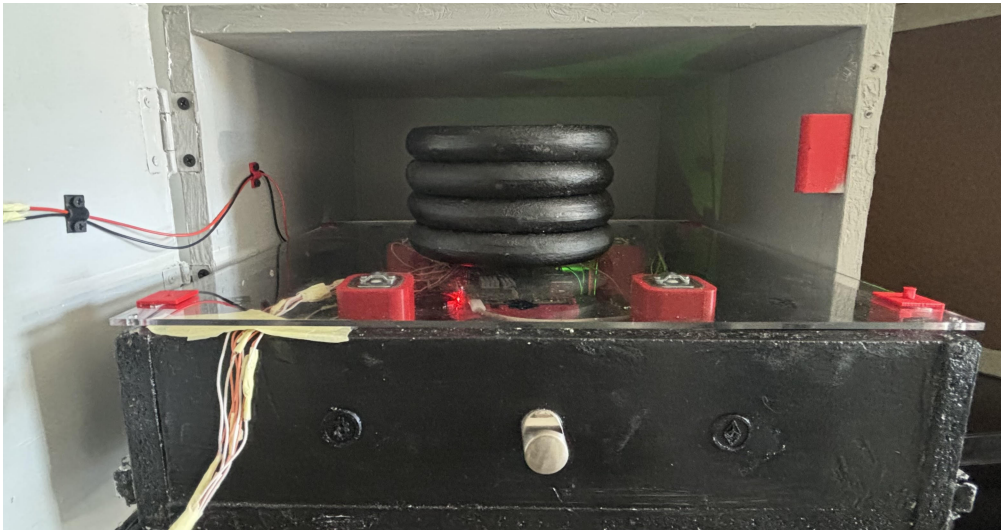


Figure C.7: 10KG 10Hour weight stress test

Appendix D

Example snippets of AI prompts

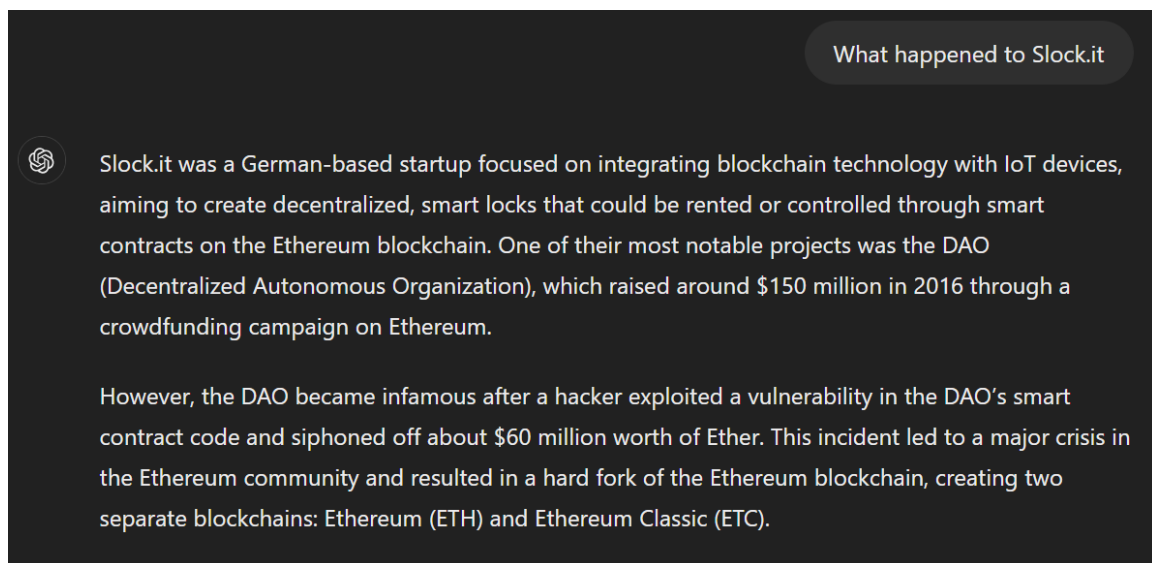


Figure D.1: Prompt example 1

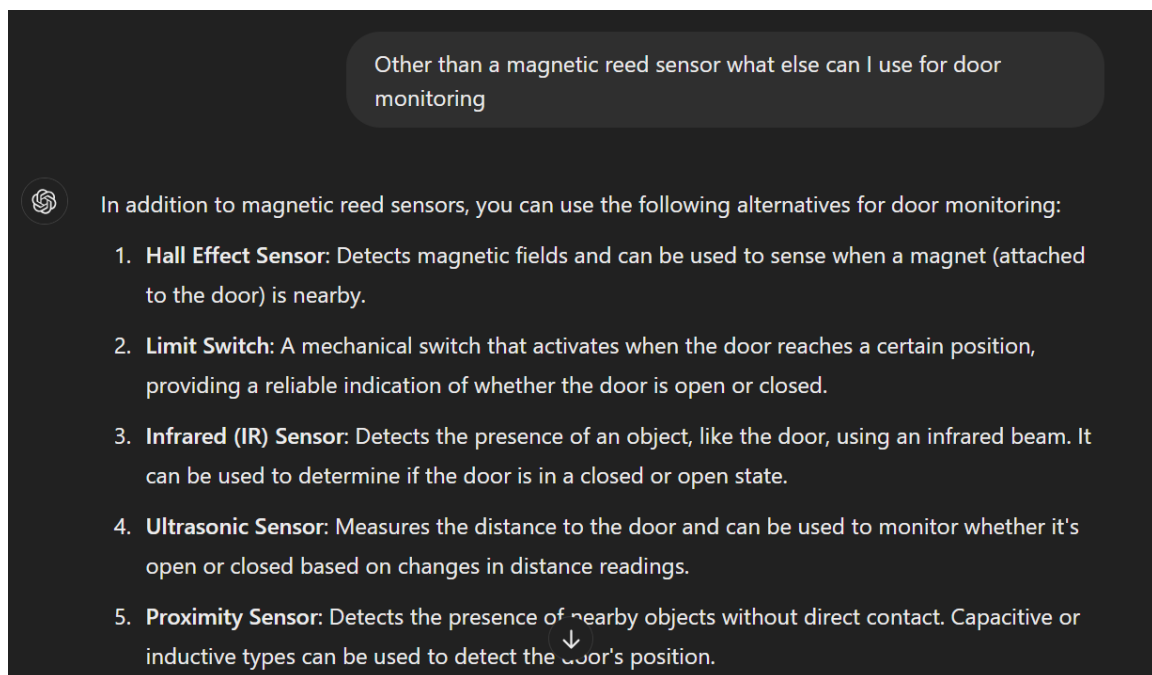


Figure D.2: Prompt example 2

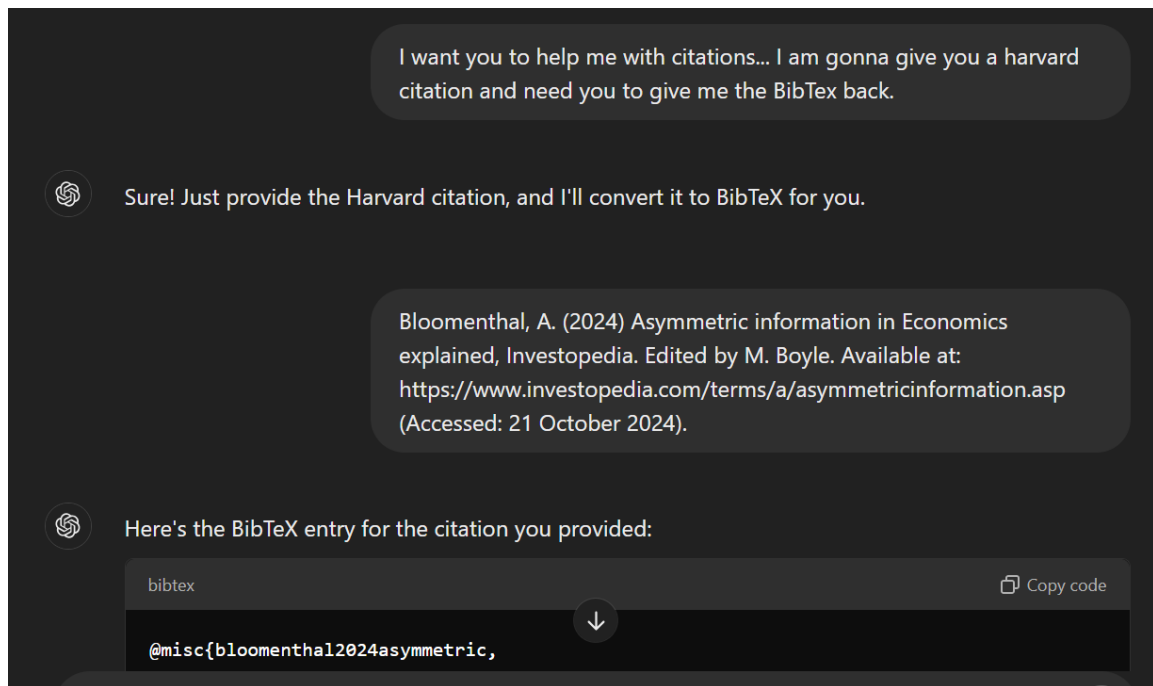


Figure D.3: Prompt example 3

Moreover, although the main back end and server code was self-developed, an LLM model helped in the coding of the Tkinter GUI.